



Unit 2: Project Planning

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Introduction to the Unit

In this unit, we will learn about the *Planning* and *Execution* phases of the project management life cycle. Upon completion of this unit, we will have a fair understanding of the various activities performed in these two phases of the project life cycle.

Some of the concepts that we will examine closely include *Project Quality Management and Value Engineering*, *Project Management Information System* and the *Work Breakdown Structure*. We will also understand the *Deliverables of Project Management*, the different types of *Organisational Structures* and the processes involved in *Estimating Activity Duration and Costs* and the importance of *Identifying Critical Activities* and *Reviews* in a project.

Objectives

Upon completion of this unit, you will be able to:

- Describe the various functions performed in the Project Planning and Execution Phases
- Identify the various types of organisational structures
- Explain the process of estimating activity duration and costs
- Describe the two popular estimation techniques—PERT and CPM
- Explain Project Quality Management and Value Engineering
- Describe the Project Management Information System
- Create a Work Breakdown Structure
- Identify the different types of reviews in a project



Organisational Structure

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Objectives

Upon completion of this segment, you will be able to:

- Describe the different types of organisational systems and the rationale for them
- Identify the various types of organisational structures
- Explain the roles and responsibilities of a *Project Manager* in each of the different types of organisational structures
- Apply the principles of a project management office (PMO)
- Rank the degree of project management maturity—of an organisation—using PMMM

Introduction

In the previous segment, we learned about the concept of portfolio management. In this segment, we will focus on understanding the different types of organisational structures and their influence on the project life cycle.

One of the emerging trends for carrying out business today is for us to focus our efforts around projects in their various forms. Management scholars are of the view that much of the work of the future will centre on knowledge generation. This will be performed in semi-permanent networks by small, project-oriented teams.

In this segment, we will examine the different types of project-based organisations and what their influence has been on the practicing project manager. We will look more closely at the role the *Project Manager* plays in each organisation form (pure project organisation, functional organisation and matrix organisation). Taking the concept of 'organisational structures' a little further, we will also discuss the concept of the project management office (PMO). Additionally, we will touch on a concept that is gaining widespread popularity, the project management maturity model (PMMM).

Project-based Organisations

Project-based organisations are firms whose primary means of doing business is by projects. Typically, these organisations are further classified into one of the following two organisational forms:



Non-project driven

These are organisations that earn revenue by *performing projects*. It is apparent that many organisations that earn revenue by performing projects often do not themselves have strong project management methodologies in place. With the lack of sufficient systems to assist with tracking and reporting, executing project management successfully proves to be quite difficult in these organisations.

Project driven

These are organisations that earn revenue by adopting *management by projects*. Organisations that are project driven have more refined management structures in place and they benefit from the *management* aspect of project management. Typically, these organisations gather information from various departments regularly and track and report on multiple concurrent projects simultaneously.

Contracting organisations, such as those in construction or consulting, earn revenue by *performing projects*. Organisations that focus on project management as the means for driving the business belong to the second category.

Organisational Forms

Project-driven organisations have a tremendous impact on the projects that they undertake. The *nature* of the organisation too impacts the project output and the *Project Manager's* approach. Now, let us look more closely at the role the *Project Manager* plays in each of these organisation forms.

The three types of organisational forms are:

Functional organisation

In a functional organisation structure, there is a hierarchical organisational structure wherein people are grouped as per their area of specialisation and are supervised by a functional manager who has expertise in the same field. There is no need for a *Project Manager* in this type of a structure.

Matrix organisation

In a matrix organization structure, there is a fine blend of, both, the pure and functional organisation structures. Here, the employees have a dual reporting structure, wherein the authority of the functional manager flows vertically downwards and that of the *Project Manager* flows sideways.

Pure (or projectised) organisation

In a projectised organisation structure, the project manager has all the power and authority and everybody directly report to the project manager. Here, either no functional manager exists, or if one exists, he/she will have a very limited role.

According to the Project Management Institute (PMI), the responsibilities of the project manager increase as a firm goes from a functional, to a matrix, to a pure, or projectised, organisational form.

Organisational Forms and the Roles of Project Managers

Let us look at the role of Project Managers in each of the organisational forms.



Functional organisation

The role of a project manager in a functional organisation:

- is akin to that of a coordinator or a team leader,
- has relatively little decision-making authority,
- the job is only a secondary requirement for the individual,
- may only be a part-time role and the *Project Manager* will devote the remainder of his or her time to core functional responsibilities.

Matrix organisation

In a matrix organisation, the role of the *Project Manager* can be that of a coordinator or of a *Project Manager*, depending on the maturity of the matrix structure.

In a weak matrix form, the *Project Manager* will, most likely, serve in the capacity of a coordinator, having limited:

- authority over project personnel and limited decision-making abilities, and
- flexibility of decision making regarding budgets.

In a balanced matrix form, the *Project Manager* will have increasingly more influence on the project.

In a strong matrix form, a *Project Manager* will more likely be a full-time *Project Manager* with a high level of authority and responsibility over the project personnel and each of the decisions that are made.

Pure (or projectised) organisation

A *Project Manager* in a pure (projectised) organisation will:

- have a full-time job, being a dedicated Project or a (multi-project) Programme Manager;
- have high authority in decision making;
- be fully responsible for all of the members of the team;
- have hire and fire responsibilities; and
- be in charge of all aspects of progress review.

The Project Management Office (PMO)

Now, let us take the concept of 'organisational structures' a little further and discuss the concept of the project management office (PMO).

Definition

A project management office (PMO) is a group formed within an organisation for the express purpose of supporting project management efforts.

Establishment

PMOs are likely to be established in:

- pure (or projectised) organisations,
- project-driven organisations, and



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- organisations that have adopted management by projects.

Composition

The personnel composition of the PMO includes:

- Project managers
- Project administrators
- Project schedulers
- Planners
- Project management software experts
- Project management support personnel (archivists, clericals, secretaries, etc.)

Most PMOs are led by a Project Director who reports to the Vice-President, or the President of a company. Under the leadership of the Project Director, subordinates may be comprised of Senior Project Managers, Project Managers, Project Administrators and Project Schedulers or Planners.

Benefits

The PMO:

- strives to support all aspects of managing projects such as training, administration, development and implementation of a set of standardised processes and procedures for managing projects;
- helps to define and promote career paths for *Project Managers* by setting standards and assisting with professional development and continuing education; and
- helps to promote success stories as well as increase the rate of project success.

Using the lessons learned through project meetings and project archives, a PMO will mature, improving its methodologies and increasing the group's effectiveness.

Responsibilities of a PMO

PMOs function as independent departments just like any other line organisation such as marketing or accounting. On the other hand, depending on the level of maturity relating to project management in the organisation, the PMO may exist within another group.

Regardless of where the PMO resides, at the conclusion of a project, most project managers operating within a PMO will feel safe going back 'home', or into their PMO, until the next project assignment comes their way.

In most cases, the PMO is partly or fully responsible for managing the project portfolio. If a PMO exists, the project managers will most likely have the appropriate level of knowledge to analyse the portfolio of projects and determine the appropriate amount of resource usage across all active and planned projects.

Additionally, using the information from the portfolio, the PMO will be able to work with the higher management to ensure that all efforts being pursued are part of the organisation's overall strategic goals.



Pros of a PMO

Templates

An example of using consistent methodologies by the PMO can be illustrated by the use of templates. When the same template(s) is used consistently from project to project, others in the organisation who have a stake in the project(s) will immediately know what to expect when reviewing reports, responsibility matrices or meeting agendas.

The use of consistent methodologies and 'boilerplates' across an organisation, coordinated by the efforts of a PMO, is particularly important when resources are shared across multiple projects as often as they are. Imagine having to learn new meeting formalities, the manner in which responsibilities on a project are documented or even rating a meeting's effectiveness in a different way every time you work on a different project.

Earned value

Another example of where consistent methodologies may come into play is in the use of earned value. The PMO may also establish the manner in which risk is analysed and mitigated, as well as the manner in which metrics or indices are used to determine a project's performance across all projects in the organisation. These metrics or indices can be unique to the firm as they fulfil its strategic vision.

Cons of a PMO

While a PMO can help to support the consistent use of a standard set of project management best practices, there can be a few disadvantages if one is not careful in setting up such an office.

Resources

Establishing a PMO requires additional resources and efforts by project managers in order to maintain consistent policies and procedures. These additional efforts, if taken to the extreme, could side-track project managers from focusing on their specific project goals.

Focus

In essence, the *Project Manager* might tend to focus on meeting strict reporting requirements in lieu of actually getting the work done.

Cost and time

Additional time and money will need to be budgeted in order to keep the PMO running, and these funds and time have to come from some place—usually from project efforts. Furthermore, with a PMO in place, additional layers of hierarchy sometimes need to be navigated by project management personnel as well as other members of the organisation.

Experience

If the PMO is not well established or respected in the organisation, members of the group may be subjected to negative experiences. Their association with the PMO may compromise functional career paths ('we' versus 'them'). Others may view the PMO as a territorial group that tries to



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impose too much control over the company's projects—a lack of freedom to get the actual work done, especially when deviations from planned effort (work) are essential.

Despite some of these caveats, a successfully established and run PMO can set one organisation or one division in an organisation apart from any other in consistently achieving successful results in the areas of project scope, time, cost, and quality.

The Project Management Maturity Model (PMMM)

Project management has gained in popularity as an organisational form in today's business arena. This, coupled with the efforts of Project Management Institute (PMI) to support the practice of project management as a true profession, has established processes to help build and assess the maturity level of project management methodologies in organisations.

Similar to other quality initiatives such as International Organization for Standardization (ISO) and Carnegie Mellon University's Software Engineering Institute's Capabilities Maturity Model (CMM), which analyses an organisation's maturity of software processes, the *project management maturity model* (PMMM) assists organisations in implementing and 'growing' best project management procedures.

The PMMM operates on the assumption that, as an organisation grows in maturity and moves from one level to the next, efficiencies are gained resulting in higher profit margins and more projects being completed on time. Ultimately, the notion is that by implementing PMMM and striving to reach level 5, an organisation will achieve a competitive advantage. Although there is little empirical evidence to support this claim, many organisations are looking for ways to reduce the risk of project failure by implementing maturity models.

A study published by the Standish Group in their CHAOS report for 2004 indicates that project success rates have increased by 34 percent. Unfortunately, the study also found that the total dollar waste in the U.S. alone was USD 55 billion with USD 17 billion resulting in cost overruns. On top of these astonishing results, these findings were only representative of the IT industry. Given these figures, it is not surprising why many organisations are implementing PMMM.

Five levels of PMMM

PMMM closely resembles the CMM in that it ranks an organisation based on five different levels of maturity. Each level, as it ascends from 1 to 5, represents a greater degree of project management maturity.

You can think of the five levels as *building blocks* or steps. As an organisation progresses in the field of project management, it builds on the cumulative knowledge of the lower level it previously attained. Typically, organisations do not skip levels. Rather, they incrementally build one level at a time.

The five PMMM levels are as follows:

Initial

This is the default level for all organisations. Policies and procedures are not well defined. Each person or project operates according to his or her own standards. Success is not achieved by implementation of best practices.



Repeatable

Basic processes are established to track cost, schedule and performance data. Project management methodologies are established to the point where cost, schedule and performance can be tracked. Projects tracking on similar tasks to predecessor projects can be implemented in a repeatable fashion.

Defined

Processes are documented and standardised across all projects. Project management best practice processes are well documented and are implemented in a consistent manner throughout the organisation.

Managed

Processes and products use quantifiable metrics. Details of the processes and quality from project to project are measurable and quantifiable.

Optimised

Continuous improvement is implemented using metrics established and re-established from level 4. Data collected are analysed to determine where efficiencies can be gained. Continuous process improvement policies are implemented consistently and thoroughly in the organisation.

PMMM Knowledge Areas

In conjunction with ranking an organisation on one of the five levels, PMMM also takes into account the level reached by an organisation in each of PMI's knowledge areas listed in the Project Management Body of Knowledge (PMBOK).

These knowledge areas are:

- Project integration
- Scope
- Time
- Cost
- Quality
- Human resource
- Communications
- Risk
- Procurement management

Each of the knowledge areas is further broken down into smaller and more refined sub-areas for additional evaluation. For instance, risk is classified into the following sub-areas:

- Risk identification
- Risk quantification
- Risk response development
- Risk control
- Risk documentation



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Some of the benefits of maturity models are based on the fact that they focus on incremental improvements based on quality initiatives. Being incremental is the key factor, which allows organisations to reach levels of maturity bit by bit. Assessments are generally easy to conduct via paper or electronic surveys. The repeatable processes that PMMM emphasises allow organisations to measure progress over a period of time.

However, there has been some criticism regarding these models. Some observers feel that the models are not robust enough to handle change. These models help the companies to identify the problems, but not necessarily to implement plans for resolution. Perhaps too much emphasis is placed on processes without respect for people or the organisational aspects.

Summary

In this segment, you learned that:

- project-based organisations are firms whose primary means of doing business is by projects;
- the three types of organisational forms are matrix organisation, functional organisation and pure (projectised) organisation;
- the responsibilities of the project manager increase as a firm goes from a functional, to a matrix, to a pure, or projectised, organisational form;
- project management office (PMO) is a group formed within an organisation for the express purpose of supporting project management efforts; and
- PPPM operates on the assumption that if it is implemented in an organisation and it strives to reach level 5 then the organisation will gain a competitive advantage.



Contemporary Organisational Issues

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3. Corporate Culture
4. Internal and External Factors
5. Global Issues
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Objectives

Upon completion of this segment, you will be able to:

- Describe the various corporate cultures that influence project management
- Describe the political pressures placed upon projects by stakeholders
- Describe the global issues impacting project success

Introduction

In the previous segment, we learned about the different organisational structures. In this segment, we will examine some of the contemporary issues that are common to most organisations.

As project management practices expand around the globe, project managers need to become more sensitive to a variety of factors that impact the manner in which projects are executed. These factors involve issues such as company culture (the spoken and unspoken rules under which businesses operate), nationality (the myriad customs and traditions followed by various ethnic groups around the world) and time zone changes from region to region.

Corporate Culture

Just as the many projects that organisations undertake are unique, so too is the 'corporate culture' under which project management is implemented.

Corporate culture can be defined by:

- the collection of beliefs, expectations and values shared by an organisation's members;
- traditions that are passed on from one generation of employees to another; and
- norms that influence suitable behaviour of employees of the organisation.

It is important for project managers to understand the culture of an organisation before trying to embark on the execution of a project. This holds true for both the organisations which formally 'own' the project as well as the organisation(s) that will contribute to the project (contractors, suppliers and even clients).

Before a *Project Manager* establishes the set of procedures that the project will follow, it is imperative that the PM makes sure that these rules are in synch with the corporate climate. A simple example of corporate culture can be understood by looking at the operating hours of employees.



While an organisation may post that normal operating hours are from 8.00 am – 5.00 pm with an hour's break for lunch, corporate culture may dictate that acceptable hours are from 7.30 am to 6.00 pm with leeway for employees to take an hour and a half break over lunch. Further, corporate policy may dictate that business casual is appropriate attire, but corporate culture may dictate that wearing denims on all week days is acceptable.

Additionally, meetings—that are scheduled to begin promptly at the time for which they are set—may be dictated by a more casual corporate culture that indicates strolling into a meeting 10 minutes late is acceptable behaviour.

Another factor that may influence a *Project Manager's* success rate has to do with understanding the role of the PM. Before embarking on the project, PMs should grasp a very clear picture of their responsibilities, authority and corporate politics. If the PM is considered a coordinator of sorts, it is important to understand to what extent their authority carries.

1. Can the PM enforce certain project rules?
2. If not, who has the ultimate authority and what are the proper protocols to follow?

Without taking the time to understand these 'unspoken' nuances of the corporate culture, project managers may find themselves frustrated by lack of respect for policies and in turn may frustrate the project team.

Internal and External Factors

Projects undergo pressures from inside the organisation as well as outside it. Most often, pressures come from the various stakeholders that have a vested interest in the project. Pressures can stem from the need to remain competitive in the market place as well as the need to please new and existing customers. Sometimes the pressure to deliver a project in a certain manner or at a certain time is dictated by these political necessities to appease the stakeholders.

Most often, these pressures are not fully understood by/or clearly communicated to the *Project Manager*. It takes a special breed to navigate the political framework of organisations to fully comprehend the constraints under which a project is managed. Project managers are sometimes forced to delicately tip toe around these constraints in order to balance the needs of the organisation as well as the end client.

Global Issues



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Another type of organisational issue—that faces project management—stems from projects spanning the globe. With technological advancements, managing virtual teams has bridged the gap of international commerce. No longer do teams need to be in the same office, let alone same country. The pool of talent available to the projects is quite vast. Project teams can operate from Asia to Europe.

While the limitation of multinational teams is being reduced by technology, project managers have new challenges to face when managing teams around the world. Such factors such as time zone issues can make coordinating deliverables an organisational challenge. Scheduling meetings with respect to multiple time zones may pose a test in creativity.

Further, allowing teams to communicate with each other in order to build team cohesiveness is yet another issue. While email has allowed teams to communicate at any time of the day, it is not always the most effective means to send instructions or status. It is sometimes necessary to communicate in real-time.

Therefore, while one team may need to get up at the crack of dawn, another may need to stay up till the wee hours of the night in order to hold a conversation via telephone or on-line chatting.

Further, not all regions of the world are on the same playing field when it comes to technology. Consideration needs to be given to internet connections and other limitations of technological advancements.

Working with international teams may necessitate face-to-face meetings, which means that travel costs will need to be built into the project's budget. Additionally, cost estimates may be influenced by currency conversion. Variability in exchange rates and tracking budget costs will require a project manager who has meticulous attention to detail. Additionally, government regulations of various countries may dictate minimum payment practices.

Even further, the maximum number of hours an employee can work per week may also be limited by government laws and or local associations such as unions. Respect to recognised holidays and vacation time is another factor that will impact a project's success. Quality standards can be a difficult issue to communicate. Expectations in one country can be vastly different from those in another.

Additionally, governments may require that certain quality standard tests be performed before an item can be released to the end users. As is the case with time being a driving factor, quality may be of utmost importance in one culture, yet a low priority in another.



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Further, language barriers can also create communication difficulties. A project manager who works with teams that span the globe needs to be an extremely effective communicator. Certain deadline driven vernacular (for example, immediately, as soon as possible and so on) can have different meanings in different cultures. Time is not always a driving force in every culture. In some cultures, the overall goal of the project is more important than getting the project completed as scheduled.

Operating autonomously with independent decision-making may not be encouraged in some cultures. In some countries getting consensus from everyone is the standard.

In summary, some of the global issues that project managers need to consider are:

- time zone variances,
- currency and exchange rates,
- communication barriers,
- respect to the variety of cultures,
- government or association regulations,
- technological limitations, and
- quality standards.

Summary

In this segment, you learned that:

- it is important for project managers to understand the culture of an organisation before trying to embark on the execution of a project;
- projects operate under various internal and external pressures, mostly from stakeholders with vested interests;
- in the era of globalisation, many projects have a global reach and accordingly give rise to many global issues which need to be managed with sensitivity.



Organising a Project

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3. Recruiting the Project Manager
4. Recruiting the Project Team
5. Organising the Project Team
6. Assigning Work Packages
7. Computer Software Applications
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Objectives

Upon completion of this segment, you will be able to:

- Describe the 'organising phase' of project management
- Describe process of recruiting the *Project Manager* and the *Project Team*
- Describe the process of organising the *Project Team*
- Explain the process of assigning work packages
- Identify the computer software applications used in a project

Introduction

In the previous segment, we learned about some of the contemporary organisational issues faced by most organisations. In this segment, we will focus on understanding the organising phase of a project. We will learn the strategies for selecting the *Project Manager* followed by those for selecting the *Project Team*. Once the *Project Team* is in place, we will see how to determine a strategy for organising the team. We will see the impact of cultural dimensions in organising our work activity.

Just like breaking the project down into smaller, more manageable pieces, we need to assign the various pieces or work packages to responsible individuals, sometimes known as project or technical leads. What are the work packages and how will the assignments be made to them? We will understand this as we discuss the process of assigning work packages.

Finally, when discussing computer software packages, we will practice assigning different resources to project activities and evaluating the consequences of making these assignments. We will also learn how to identify if a resource is overloaded and how to manage such resource usage.

Recruiting the Project Manager

Recruiting the *Project Manager* can be a complex task. The *Project Manager* is the focal point—the person who is *ultimately responsible* for the project's success or failure. The *Project Manager* must be a strong leader, yet a diplomatic negotiator. He or she must be disciplined enough to enforce processes, yet flexible enough to allow for changes that inevitably occur in all projects.



When to hire a project manager

Ideally, recruitment of the *Project Manager* should begin during planning, or very early in the defining phase of the project, or alternatively, as soon as possible. The earlier a *Project Manager* is identified, the greater the advantage he or she will have in getting up-to-speed in managing the project. If a *Project Manager* is identified during the planning phase, he or she will be able to participate in development of the project proposal and become more intimately familiar with project budgets, resource requirements, negotiation of the project scope and the type of contractors and other external resources that will be used.

Additionally, he or she will most likely have access to internal project contacts and fully understand the constraints that must be considered in planning the project, and throughout the project's duration. Further, if the project manager is recruited early on in the project development phase, he or she may be able to begin forming relationships with key stakeholders (for example, the various clients and the executive management). Having a solid relationship with the client will help the project run more smoothly throughout, especially when adjustments have to be made.

Project Manager – An afterthought

Unfortunately, not all projects run according to our 'ideal world'. Frequently, project managers are afterthoughts. Management may not even begin to think about the *Project Manager* until after a project has begun or a deliverable has been promised to a client. The focus has been on *what* must be done rather than *who* is going to do it.

As we have already seen, many projects are selected when the executive management asks an individual to 'look into something', also known as the 'sacred cow' (a type of project selection model). Management may be thinking of one person to plan the project, but yet another to lead or direct it. Or, they might not even have thought ahead sufficiently for this phase or who will direct the effort.

Instructions must always be clear, but especially when an individual sets off to accomplish a project without initially identifying a *Project Manager* to lead it. A more favourable situation almost always exists when the individual asked to 'look into' a project is also the *Project Manager*. In other cases, it might be difficult to assign a leader before work begins on a project, due to the difficulty in finding a suitable *Project Manager*.

Additionally, due to turnover or reassignment, project managers may come onto, or be removed from, a project at various phases or milestones throughout its life. The textbook, Meredith, J. R. and S. J. Mantel, *Project Management: A Managerial Approach*. 6th ed. John Wiley and Sons, 2006, discusses a situation where a chemist on a project serves as the *Project Manager* during the beginning phases, and then a marketer assumes the *Project Manager* role during the later stages of the project lifecycle. The organisational structure and processes that are established within the organisation will ultimately dictate recruitment of the *Project Manager*.

The drawbacks of hiring a project manager late

Learning curve

One obvious disadvantage of assigning a project manager at a later stage in the project lifecycle is the learning curve the *Project Manager* will face. He or she will have to spend significant time gaining familiarity with the project (that is, reading through contracts, understanding the project



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scope, deciphering the project culture and establishing himself or herself as the leader and gaining respect from team members).

Relationship management

The *Project Manager* will have to build a relationship with the client. If a *Project Manager* is assigned to a project to replace another who has left, the relationship management may be even more difficult due to previous, established processes with which the client is familiar. Those assuming the later leadership role must be aware of the sensitivity of the situation, and the requirement for early communication with everyone involved early in the management or leadership of the project.

Ideally, in this situation—which is not the most favourable—the departing *Project Manager* can introduce (through a variety of mechanisms) the replacing individual to all the important stakeholders including, obviously, the *Project Team*. As much overlap as possible between incoming and outgoing project managers is always advisable.

When recruiting the *Project Manager*, several criteria regarding skills-set must be considered. Depending on the perceived level of project difficulty, management must analyse the level of industry and technical expertise required for the project.

Managing a space shuttle launch for the National Aeronautical and Space Administration (NASA) of the US requires a *Project Manager* with a different skillset than someone charged with the duty of constructing a new highway or developing an entire complement of microwave oven products for a home appliance manufacturer. Conversely, someone who understands the nuances of organising a conference may be well suited to manage a company's 100th year celebration.

In general, project managers should have the following attributes:

- appropriate level of expertise to anticipate problems and identify contingency plans before problems become fully developed;
- appropriate level of industry knowledge for the project;
- ability to switch from being a specialist to being a generalist in order to look at the project from a bird's eye view and integrate all of the pieces of the puzzle to form one cohesive and successful effort; and
- passionate about the project assigned to them, enthusiastic almost to a fault—and become the champion or the main spokesperson for it.

Skill set that a project manager should possess

Leadership skills

Management must also consider if the candidate is capable of leading a team of 50, 100 or possibly even 1,000 team members. Will the *Project Manager* have the appropriate level of interpersonal skills to manage the various personalities that will comprise a team? Management must be sure that the team will accept and recognise the assigned person as their leader. Furthermore, he or she must have excellent communication skills to ensure that all team members understand and commit to the project objectives.

The *Project Manager* must be able to 'rally the troops' when things get tough. They must promote team spirit and assist others in navigating complex issues. They must be able to identify which tasks are best delegated to others and which tasks must be managed single-handedly.



If the project requires the use of outside contractors, management will need to consider if the *Project Manager* has the appropriate level of leadership skills to assimilate the contracted resources with other internal team members and co-ordinate the efforts between them.

Part of the leadership skill is the ability to make sound decisions. The *Project Manager* must be able to obtain data from converging areas and filter the appropriate information to formulate a solution approach.

Communication skills

A *Project Manager* must possess effective communication skills, or he or she is unlikely to get the job done. Without sufficient communication, tasks cannot be completed, unless through luck. And luck is never a substitute for effective management or organisation in a project (although sometimes it helps). The *Project Manager* is responsible to the organisation, the client, his/her team members and other important stakeholders. He/she must ensure that the appropriate project details are communicated to each stakeholder group.

The following questions should be addressed first-hand by the *Project Manager*:

1. How frequently are meetings going to take place and when?
2. Where shall the participants meet?
3. How far in advance will the agenda for the meeting be communicated to the involved participants, and how shall it be communicated?
4. What will be the format of the project update meetings? Who gets to say what and when?
5. How much time will be allowed for each participant to talk at project update meetings?
6. Who will take notes during the meetings, and how shall these notes be distributed to the participants and when?
7. Where shall the log of meetings and updates reside, and who shall have access to them?
8. How many copies of the meeting logs shall be made available?

Presentation skills

Another important attribute to consider in effective communication is the *Project Manager's* presentations skills. Will he or she be able to deliver information effectively to the *Project Team* and in an equal manner to each participant? Will he or she present effectively to the upper management and other important stakeholders?

Organisational skills

The *Project Manager* should also have excellent organisational skills. He or she should put project standards and methodologies into place, and should clearly articulate project expectations so that all team members and stakeholders understand their various roles and responsibilities.

The *Project Manager* may even consider developing a responsibility matrix to assist in communicating these expectations. With the matrix in place, key project players can clearly see who to go to for what and identify for which items they are accountable.

Further, the *Project Manager* should think big enough to plan the project effort at a high level, as well as be detail oriented enough to not miss important project requirements. The *Project Manager* should ensure that his or her project goals are in line with the company objectives without compromising client expectations.



Administrative skills

The *Project Manager* should also be able to implement various administrative functions required in successful project completion. Whether the *Project Manager* is responsible for carrying these out single-handedly or has support staff such as a project administrator, they should be able to control such tasks.

The *Project Manager* should ensure that:

- agendas are published before all meetings,
- minutes are made available to all stakeholders in a timely manner,
- budgets and schedules are updated and maintained,
- invoices are sent according to the project terms and conditions, and
- reports are prepared and shared with the appropriate audience.

For all these to happen, the *Project Manager* should possess excellent communication skills.

Other skills

Apart from the above skills, a *Project Manager* is also expected to:

- handle conflicts and resolve them,
- negotiate deals, and
- deal with cultural aspects of the project.

Not only will the project and its team have a culture of its own, cultural influences exist within the organisation and between the organisation and its various stakeholders. Some of the cultural issues are:

- What are the reporting hours?
- What are local customs?
- Is overtime allowed, and if so, how are overtime schedules set?
- What is the protocol concerning communication between superiors and subordinates?
- What form is the communication to take in a project?
- What is the dress code?

The dictionary defines culture as ‘the total pattern of human behaviour and its products embodied in thought, speech, action and artefacts.’ Culture is dependent on the capacity for learning and transmitting knowledge to succeeding generations using tools, language and systems of abstract thought. Culture involves the body of customary beliefs, social norms and material traits constituting a distinct, complex tradition for its members.

No matter how much we understand about another’s culture, especially when participation involves team members across international boundaries, our general understanding is just the tip of the cultural iceberg.

The cultural iceberg

As seen in the cultural iceberg below, no matter what we *think* we understand about working with a different person and the influence of his or her culture, we have only scratched the surface in understanding all we need to know about the person and how culture influences so much of the



their behaviour. A well-grounded understanding of culture goes a long way towards developing an understanding of a person's behaviour and why it differs from ours.

Culture - It may not be what you think it is





Characteristics – in a nutshell

The following list identifies some of the characteristics, which should be considered when recruiting a *Project Manager*, to ensure that he or she will be a good fit for the project.

No.	Characteristics of a Project Manager
1	Industry Expertise
2	Technical Expertise
3	Strategic Thinking Ability
4	Leadership Ability
5	Analytical Skills
6	Motivational Skills
7	Conflict Resolution Skills
8	Negotiation Skills
9	Decision-making Skills
10	Organisational Skills
11	Planning Capability
12	Ability to Facilitate and Delegate
13	People Skills
14	Communication Skills

Costing factors to consider when hiring a *Project Manager*

Last, but not least, management should consider the cost of the individual assigned to direct the effort. The following questions should be addressed:

- Will this person be a contractor, consultant, contract-to-hire or a recruit from within?
- Can the project's budget absorb the cost of the new hire?
- Are the skills lacking internally, for a person to direct this effort? If so, what shall be the mechanism for obtaining the services of an outside individual?
- What are the plans for this individual upon completion of the project?

Further, management must look at their overall strategic goals to determine if hiring a full-time *Project Manager* is in line with their organisation's growth model or if recruiting internally makes the most strategic sense.

Recruiting the Project Team

In order to get the project off the ground, a team of resources must be assigned to the project. The group of individuals who take part in determining who will comprise the team may range from executive board members to functional managers or, sometimes, just the *Project Manager* working alone. As with recruitment of the *Project Manager*, the selection process of the team will most likely be influenced by the company's organisational structure and policies.

Determining the number of resources

One of the first items to consider when assigning the team to a project is the quantity and type of resources required in order to meet the project objectives and schedule. In conjunction with determining the quantity of resources required, the *Project Manager* also has to take into consideration the project budget.



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If the number of resources required to meet the schedule objectives exceeds the project budget, the *Project Manager* has to reassess the current plan. He or she may attempt to renegotiate the project terms and conditions with the client, implement creative planning techniques, such as project crashing and project fast tracking, or see if alternative resources, with a more advanced level of expertise, can be obtained to expedite the schedule or meet the costs or other objectives.

Identifying the appropriate resources

In reference to creative planning techniques, team skill sets will first need to be analysed.

1. If the project to be undertaken is similar to projects that have been executed in the past, it is likely that the same or similar resources can be used on the current project.
2. If the project is a new undertaking or attempts to use a new technology, alternative resources may have to be sought.
3. Should a company be fortunate enough to have the appropriate level of expertise within the organisation, the project manager (or decision-making body) will need to determine if the resource(s) are available.

When identifying resources to work on a project, the team skills set has to be analysed and considerations have to be made to match these with the requirements of the project. Then, we need to go out and recruit the right individuals for the project.

Recruiting the right resources

Some of the characteristics of an individual that a *Project Manager* would consider when recruiting team members are as follows:

- prior experience of working on similar projects;
- prior experience of working with the existing team members;
- prior experience of working with the same client;
- prior experience of undertaking a similar role;
- right type of personality to mix with the rest of the team;
- right type of personality for the role;
- ability to bring new skills on-board;
- ability to add creative flare; and
- communication, presentation and writing skills.

For example, if a *Project Manager* is recruiting a team leader, he or she is likely to look for someone with leadership qualities and prior experience of managing teams. On the other hand, if a *Project Manager* is looking for say a designer, he or she may look for someone who is able to add creative flare to the project.

Taking recruitment decisions

If you were the *Project Manager*, how would you recruit a project team? How would you source for the team and from where would you procure your resources?

As in recruiting a *Project Manager*, consideration will also have to be given to the appropriate fit of resources assigned with the rest of the *Project Team*. Additionally, the *Project Manager* may need to negotiate with functional managers for the resources necessary to work on the project. The *Project Manager* or the decision-making body may need to make concessions regarding personnel



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recruitment policy depending on project constraints, or constraints imposed by other on-going, and possibly higher-priority projects.

The time required to recruit a suitable project team should be considered when planning the project schedule and estimating project completion time. In some cases, interviewing may be required. Candidates recruited from outside may have relocation issues to consider (selling or buying a home, logistics of moving, etc.)

In some cases, depending on the mix of resources already assigned, it may be necessary to start on the project while the remaining project members are being recruited. If the *Project Manager* feels that external contractors will benefit the team, he or she may need to convince the *Project Stakeholders* of the merits of this approach in order to gain their approval. Once the use of external resources is approved, the *Project Manager* has to negotiate with the outside individuals regarding their terms and conditions.

Working with sub-contractors usually adds another layer of administrative duties for the *Project Manager*, though the difficulty of these duties depends once again on the organisational structure in the firm. Does a separate office in the organisation exist to assist the *Project Manager* with writing and advertising contract specifications and with subsequent contract negotiations, for example?

Organising the Project Team

Once key, as well as other, resources have been identified for the project, and have been assigned work package completion details, it is necessary to organise the *Project Team*. In this segment, we offer three structures – *pure project*, *functional* and *matrix* – for organising the team and indicate the advantages and disadvantages of each. The various organisational structures often evolve as a firm grows, adding resources and people to accomplish new tasks or additional ones (alike or similar activities).

It is possible that a firm can be organised by several of these methods, at different levels in the organisation, at the same time. For example, the firm might be organised functionally at the highest levels of the organisation, and be arranged by the project to accomplish the current projects undergoing completion at the middle and lower levels of the hierarchy.

In order to get the project off the ground, a team of resources must also be assigned. The group of individuals who take part in determining who will comprise the team may range from executive board members to functional managers or, sometimes, just the *Project Manager* working alone. As with recruitment of the *Project Manager*, the selection process of the team will most likely be influenced by the company's existing organisational structure and policies.

First steps in bringing the *Project Team* together

Key issues to consider when organising a *Project Team*:

Developing project plans

This may include:

- communications plan,
- risk-management plan,
- change-management plan,



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- scope-management plan,
- cost-management plan,
- quality-management plan,
- procurement-management plan, and
- staffing-management plan.

Holding a kick-off meeting

The final step in this process is to hold a project kick-off meeting to discuss the contents of the project plan. It needs to be ensured that this meeting, and all subsequent meetings, are facilitated with respect to:

- creating an open and friendly atmosphere,
- publishing an agenda prior to the meeting,
- starting on time and ending on time,
- keeping the team focused on the meeting's objectives,
- assigning action items to named team members and ensuring they accept the task being assigned, and
- assessing the effectiveness of the meeting in achieving the project goals.

Organisational structures

This section explains the three organisational forms—*pure project*, *functional* and *matrix*—and how the nature of these structures affects the way a project is handled internally. We will also identify the advantages and disadvantages of each approach.

Organisational structures and their impact on projects

Functional organisation

At the other end of the spectrum is a functional organisation structure where we have functional projects housed within the various functional divisions in the firm.

Matrix organisation

The matrix organisation attempts to blend properties of the pure project and the functional organisational form. We often find, both, weak matrix and strong matrix forms in practice, depending upon a firm's strategic priorities, among other things to consider when adopting this structure.

Each project in this organisational form involves people from various functional areas. Most often, the *Project Manager* decides which tasks to perform and when, and the functional managers determine which people and which other functional resources will be used.

In this organisational form, it is especially important that the *Project Manager* be a skilled negotiator and an excellent communicator, who can that proper resources and the proper mix of people (individuals) are available for the project work, when needed.

Pure project organisation

Tom Peters predicts that most of the world's future work will be 'brainwork', done in semi-permanent networks of small, project-oriented teams. Each team is an autonomous, entrepreneurial centre of opportunity, where the necessity for speed and flexibility renders obsolete the hierarchical



management structures of the past. The pure project organisation affords us the opportunity to respond rapidly to changing market conditions in order to take advantage of an opportunity, or to rapidly solve a problem. Team members are focused more on solving a problem at hand, rather than participating in a function.

Often, many of the functional obstacles are removed in a pure project organisation, and effort concentrates instead on meeting deadlines or rapidly solving problems. The pure project method of organising is often called 'skunk works'. This term is used a great deal in the automotive industry in developing new automobiles and prototypes.

Organisational structures and projects: advantages and disadvantages

Advantages of a pure project organisation

- The *Project Manager* has full authority over all aspects of the project.
- Team members report to one boss—there is no question of loyalty.
- Decisions are made quickly—communications are simpler than with other organisation forms.
- Team members have more of a problem-solving approach, rather than a functional orientation.
- Often 'teams' of integrated expertise develop, which is necessary for solving certain problems because any one area cannot do it alone.
- Team pride, motivation and commitment are high.

Disadvantages of a pure project organisation

- There is often a duplication of resources, especially staff. Resources are often not adequately shared among projects. Project managers have a tendency to stockpile or 'squirrel away' key resources, even when not in use.
- Organisation (management) leadership may feel less in control.
- The organisation may fall behind in knowledge and technology, as there are fewer members of pure functional divisions with which to work and share information.
- 'Life after project' anxieties can be highest among team members.

Despite these disadvantages, the pure project method of organising is often felt to be the preferred approach when speed and flexibility are the key issues. In addition, this form also enhances the entrepreneurial spirit of the team members.

Advantages of a functional organisation

- Team members are easily shared among projects. This contributes a great deal of flexibility in the use of functional expertise.
- Team members report to one functional boss—there is no question of loyalty.
- Technical expertise is maintained within a functional unit. There is usually a strong, discipline-oriented knowledge base available. True experts can be shared among projects.
- The functional area is 'home' after the project is completed. End of project life anxieties are generally lower in this organisation form.
- A critical mass of specialised functional area expertise is maintained.



Disadvantages of functional organisation

- Team members tend to have more of a discipline-oriented rather than a problem-oriented approach. This is often called a 'tool in search of a problem'. The project or the client might not be the focus, but rather, the functional unit might be too much the focus of activity.
- This arrangement does not facilitate a holistic approach to managing a project. Often times, design really suffers if the project is not considered as a whole, especially when many functional units need to be involved (as is the case in most large projects). The total turns out to be less than the sum of
- the parts due to local or sub-optimisation.
- Motivation for project work may be low.
- Response and communications to the client can be slower and fractionalised.

Advantages of a matrix organisation

- The project is generally the focus of emphasis.
- The *Project Manager* is held responsible for all aspects of the project.
- Duplication of resources is minimised.
- The project has a wealth of resources, including technical and other expertise to draw on.
- The functional units can supply capacity or individuals to the project.
- Communications between functional divisions are enhanced.
- Team members have a functional home after project completion. This leads to less anxiety over project completion.
- Policies and strategies of the parent organisation are followed.

Disadvantages of a matrix organisation

- There are two bosses, and team members (especially in a time of conflict) may support their ultimate boss, the functional unit head.
- Political infighting among projects might kill an individual project.
- The project may be doomed to failure if the *Project Manager* is not a skilled negotiator.
- The opportunity for conflict is much greater than with other organisation forms.

Regardless of which organisation structure is ultimately adopted, it is important to keep in mind that the *Project Manager* is still pivotal in successful project completion, and is the primary contact point for the client and other important stakeholders. Flexibility and responsiveness are enhanced because one person is in charge and is ultimately responsible for successful project completion.

Conflict Resolution

Certain organisational structures potentially give rise to more conflict than do other forms, when managing a project. When conflict arises, the *Project Manager* must be very skilled not only at negotiations, but also at conflict resolution. One of the ways in which conflict among team members can be resolved is through the use of an escalation process.

The guidelines to follow while resolving conflicts are:

- First, allow members who have a conflict to attempt to sort out the issue amongst themselves.
- If the attempt produces nothing but frustration, the next step is to involve the *Project Manager*.



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- If the *Project Manager* cannot facilitate a resolution, then the next step is to involve the relevant functional managers of the team members involved in the conflict, along with the *Project Manager*. The hope here is that someone with more authority will be able to make the appropriate decisions to resolve the conflict to everyone's satisfaction.
- If the attempt for resolution is again a failure, the issue is then elevated to the next authoritative rung on the ladder.
- The issue is 'escalated' through each rank of authority (all the way up to the CEO, if necessary) until the issue is either resolved or a decision is made by a ranking official. It should be noted that an issue should only be escalated after those involved have seriously tried to resolve the problem at the lower level.

The escalation process – an example

Suzy and Hans have been working on a software development project that is responsible for building their company's Enterprise Resource Planning (ERP) system. The organisational makeup for this project is that of a matrix structure. Suzy and Hans are both software developers working on the same phase of the project, but report to different functional managers.

Suzy has suggested a particular design for a component of the project, but Hans does not like it. He has suggested an alternative design for the same component. Suzy feels that her design is superior to Hans' and, similarly, Hans feels his plan is superior to Suzy's.

The *Project Manager* gets involved to help them resolve this issue. After discussion turns to arguments, the *Project Manager* wraps up the meeting by suggesting that the issue be escalated to Suzy's and Hans' supervisors, Joe and Kathy, respectively. Unfortunately, even with Joe and Kathy's involvement, the conflict does not get resolved.

The problem gets escalated to the highest level of impartial authority. The Vice President of technology, Joe and Kathy's boss, is called in to review the issue. Having taken the time to go through and understand the details of the issue, he reviews the designs, and decides that Hans' design should be used. The decision is communicated to all involved parties and the plan is implemented.

When escalating an issue through the various ranks of authority, it is important to note that as the issue gets routed to higher and higher ranks, executive management availability becomes less and less. Should the issue be escalated to someone at an executive level, the *Project Manager* and other involved parties should feel comfortable that the issue was important enough to be reviewed by someone who has very little time.

Assigning Work Packages

With the team selected, and the organisation firmly in place, the *Project Manager* can begin making the appropriate personnel assignments for each project task or work package that needs to be completed. Consideration of activity sequencing and resource availability will also need to be taken into account. The use of scheduling software such as Primavera P3E/c can assist with determining the appropriate level of resources and when to utilise them.

Not only does the *Project Manager* need to be concerned with the resource that will be responsible for executing each activity, but he/she should also consider assigning a lead person responsible for each work package. The larger a project, the more difficulty the *Project Manager* will have in managing each activity. The assignment of a lead person for each work package will allow the *Project Manager* to interact with fewer people when gauging the overall status of the project.



If the *Project Manager* is working in a matrix type organisation, the functional manager for a particular resource may be assigned as the lead person responsible for his or her department's set of work packages. Whatever method is applied, for assigning 'work package leads,' needs to fit in with the company's overall structure and methodology.

Work package assignment

Once each team member has been assigned to the various project tasks and work packages, the *Project Manager* should ensure that everyone's roles and responsibilities are clearly defined and accepted by each team member. Without buy-in by each team member on his or her accountability, the *Project Manager* will most likely have to fight an up-hill battle, throughout the project, regarding team expectations. Sign-offs on a project charter and at the work package level are particularly helpful in this regard.

Another item that the *Project Manager* needs to consider when assigning personnel to each activity or work package element is the amount of time they can dedicate to the project. This becomes a particularly important issue when working in a *functional* or *matrix* type organisation. Resources may be expected to work on multiple projects simultaneously.

Assigning personnel in functional and matrix organisations

Functional organisation

Consider an organisation that has decided to implement a new accounting system. The organisation has adopted the functional organisational approach. The accounting department has been assigned this project. While various team members from the accounting department have been assigned to this new project, they still have their day-to-day responsibilities that need to be carried out.

In this scenario, each team member has been told that they are to split their time 50/50 between the new project and their current responsibilities. Therefore, the maximum hours per day that any one team-member can spend on this project will be four hours per day (assuming that a normal work day is made up of eight hours).

Matrix organisation

Similar to the functional organisational structure, in a matrix organisational structure, multiple resources from various disciplines may also be working on several projects simultaneously. Co-ordinating team availability across different projects and functional groups can become quite cumbersome; therefore, planning is essential.

While manual-tracking techniques can be put into place, the more projects and resources involved, the more difficult implementing a shared resource pool becomes. Using a scheduling software package such as P3E/c application that allows for 'enterprise' resource planning can eliminate a lot of the effort involved in using shared resources across multiple projects.

Other Issues

Other issues to consider, when making personnel assignments, have to do with company schedules. The following questions should be asked:

1. Does the organisation use a one-shift or two-shift schedule?



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2. Do employees work a five-day work week, or do they work four 10-hour days, for instance?
3. Which holidays are observed by the organisation?
4. Are all members of the team full-time employees, or are some team members part-time employees?
5. How much productivity is expected from each team member?
6. Is anyone on the team new to the organisation, and still 'ramping-up' so to speak, and, therefore, will have to be allowed extra time to complete the work which has been assigned?
7. Is there a new skill set that needs to be obtained before being able to contribute to the project?

Calendars introduced in the next topic facilitate scheduling resources across a variety of conditions. As you can see, making appropriate personnel assignments is no easy task. However, with thorough planning and careful consideration of the numerous project variables, successful designation of resource assignments can be accomplished.

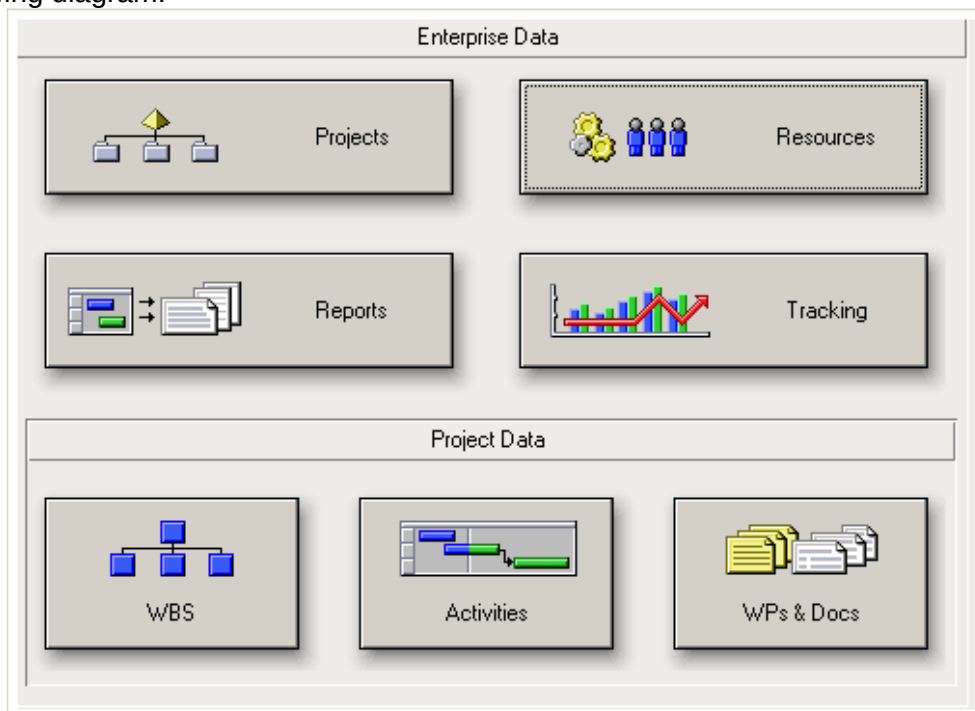
Computer Software Applications

Now that you have a good idea of what must be considered when recruiting the project manager and the project team, as well as the points to consider when making personnel assignments, we will now learn how to implement planning software such as P3E/c when organising a project and making individual personnel assignments to work packages.

Although in this topic we refer to Primavera P3E/c as an example of a project management tool, you should understand how these objectives relate to the specific project management tool you have chosen to use.

Using P3E/c

P3E/c clearly delineates the distinction between enterprise data and individual project data, as seen in the following diagram.





Enterprise data includes project collections, firm or enterprise resources, reports and the various tracking mechanisms.

Project data includes the work breakdown structure (WBS), activities and reference documents (for example, blueprints and drawings, as well as other reports.)

This distinction between firm 'matters' and project 'matters' goes a long way in helping us decide how the project should be organised in the firm and—once organised—how it should be staffed and managed in the enterprise.

Creating a New Project

For this topic, we will be entering information for establishing a project management office (PMO) that has just been created. The new PMO has been tasked with creating a project management handbook so that everyone in the group, as well as any new members that are added in the future, will consistently implement the developed project management methodologies throughout the organisation.

Creating a new project

In this simulation, we will create a new project:

1. Open **P3E/c**
2. Click **Create New...** to create a new project
3. The **Create a New Project wizard** opens
4. Click the ... button to select the Enterprise Project Structure (EPS)
5. Click the '**cross**' icon on the right of the 'Select EPS to add into' window, to select the EPS
6. Click **Next** to continue
7. Leave the default as **Project ID**
8. Enter the project name
9. Click **Next** to continue
10. Click ... from the **Planned Start** selection field to open the calendar
11. Navigate the calendar to the month January, 2005
12. Click **Select** to set 10 January 2005 as the project start date
13. Click ... from the **Must Finish By** selection field to open the calendar
14. Navigate the calendar to the month February, 2005
15. Click **Select** to set 19 January 2005 as the project end date
16. Click **Next** to continue
17. Leave the default option for the **Responsible Manager** field and click **Next** to continue
18. Click **Finish** to close the wizard
19. The project activity sheet appears to indicate that the new project has been created

Entering resources

In this simulation, we will enter new resource data into the P3E/c application.

1. Click open **File** menu
2. Click **Close All**
3. Click **Yes** to confirm
4. The main window appears



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5. Click **Resources**
6. The **Resources** screen appears with the existing resources
7. Move to the far right corner of the screen
8. Click the first icon on the far right of the screen to add a new resource
9. The **New Resource Wizard** opens
10. Enter **Resource ID**
11. Enter **Resource Name**
12. Click **Next** to continue
13. In the next seven steps, leave the default option and go on clicking **Next** till you reach the **Progress Reporter Setup screen**
14. Select the '**No, I will not set up the Progress Reporter at this time**' radio button
15. Click **Next** to continue
16. Click **Finish** to exit the wizard
17. Repeat steps 7 to 16 to add every resource listed in the table. Once all of the resources are added, the application produces a resource group for the PMO, as shown in the next screen.

Creating a Calendar

Now that we have entered the new resources, we will need to consider the type of calendar that is used by the organisation. We need to determine if the organisation follows a typical five-day work week or if it follows something a little different such as four 10-hour days.

P3E/c application comes with a variety of work calendars that can be used. It is also possible to modify or create an entirely new calendar to suit the project or organisation's needs.

In this simulation, we will create a calendar.

1. Click open the **Enterprise** menu
2. Select **Calendars...**
3. The Calendars dialog box opens
4. The window displays various 'Global' calendars that can be used in P3E/c. Selecting the 'Resource' or 'Project' radio button enables you to apply/create calendars that are specific to a particular resource or project.
5. Suppose we want to create a 'summer' calendar for the months of June, July and August, where all employees are entitled to have every Friday off (with pay). Follow the next steps on how to do this.
6. Select **Corps Standard** checkbox
7. Click **Modify** to change the calendar
8. Click **Fri** to highlight all Fridays
9. Click **Nonwork**
10. All Fridays are marked as nonworking days
11. Repeat this process for July and August
12. Click **OK** to save the settings
13. Click **Close** to exit the calendar setting

Personnel Assignments

With the resources for the PMO set up and the new 'summer schedule' created, appropriate resource assignments can be made.



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Table 2 presents the activities, durations and the predecessor/successor relationships for the example project. Be sure you set up the appropriate WBS elements before you begin entering activity information.

Table 2: Activities, Durations and the Predecessor/Successor Relationships

ID	Activity	Duration	Predecessors
A1140	Project Begins	0 (Milestone)	None
A1080	Discuss Content and Objectives for Handbook	1 day	None
A1020	Draft Subcontractor Policy	7 days	A1080
A1030	Draft Communications Policy	7 days	A1080
A1040	Draft Kick-off Meeting Guidelines	4 days	A1080
A1050	Draft WBS Development Guidelines	7 days	A1080
A1060	Draft Project Review Guidelines	4 days	A1080
A1090	Review Draft with Team	1 day	A1080
A1100	Implement Necessary Changes	2 days	A1020, A1030, A1040, A1050, A1060
A1100	Review Revised Document for Accuracy	1 day	A1100
A1120	Obtain Approval from Executive Management	5 days	A1100
A1130	Publish and Implement Handbook	1 day	A1120
A1150	Project Complete	0 (Milestone)	A1130

Assigning Personnel

The following simulation shows how the information from the above table is used to assign resources for this project.

1. Move to the far right corner of the screen
2. Click the first icon on the far right of the screen, the **Add** icon, to add a new activity
3. Leave the default as Activity ID
4. Enter the activity name
5. Click **Next** to continue
6. Click the ... button from the **WBS** selection field to select the relevant WBS
7. Select **Approval & Implementation**
8. Click the second button on the right of the **Select WBS** window to select
9. Click **Next** to continue
10. Click **Finish** to close the **New Activity** wizard
11. Now we are finally ready to assign the appropriate resources to each activity. Make sure that the 'Resources' column is displayed. You can do this by right-clicking on one of the columns and then



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selecting 'Resources' from the list of available options. You will also want to make sure that the 'Activity Detail' is displayed in the lower portion of the screen. To do this, go to the menu bar, click 'View' and then 'Bottom View'. Select 'Activity Detail'.

12. Click **Resources** tab
13. Click **Add Resource** button
14. Click to select the first resource
15. For more names, hold shift key and click to select
16. Click the **select** button (the second button on the right of the **Assign Resources** window) to assign the selected resources
17. Click **Close** to close the **Assign Resources** window
18. Follow steps 12 to 17 to assign resources to each activity.

Having come to an end of this segment, you have seen, via the simulations, how to make use of computer software to organise your project and assign resources. You have learnt how to add resources to the 'enterprise library,' make use of the various calendars such as global, project and resource-specific calendars, as well as how to assign resources to various activities.

Summary

In this segment, you learned that:

- a *Project Manager* must have a set of key traits that are necessary for the success of a project;
- recruiting and organising the *Project Team* are important steps in the project organising phase;
- the various organisational structures have advantages and disadvantages which have a bearing on how a project is organised;
- there are many factors that affect the work package assignment, such as the organisational structure; and
- Primavera P3E/c is an example of a project management tool, which is used for organising a project and making individual personnel assignments to work packages.



Estimating Activity Duration and Costs

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1. Objectives
2. Introduction
3. Determining Estimates
4. Factors Affecting Time and Cost Estimates
5. Estimation Techniques
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7. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the need for estimating activity durations accurately
- Define the process and methods available for making accurate estimations
- List the factors that affect the estimation process
- Define PERT and CPM

Introduction

In the previous segment, we understood the organising phase of the project management life cycle with special emphasis on recruitment of the *Project Manager*, recruitment and organisation of the *Project Team*, assignment of work packages and understanding how computer software applications are used in project management.

In this segment, we will understand the need for, and the process of, estimating activity durations. We will also touch upon the two estimation techniques: Program Evaluation Review Technique (PERT) and Critical Path Method (CPM).

Determining Estimates

Before we discuss time and cost estimation procedures, it is important to understand how an organisation's structure can affect the manner in which time and cost estimates are made. The two main strategies used to determine estimates are the:

- *top-down* approach
- *bottom-up* approach

Top-down approach

The top-down approach (also referred to as analogous estimating) engages the expertise of middle and upper management to determine overall project estimates. When this strategy is used, the project team is challenged to fit their activity and cost estimates into the overall project budget or project scope. When various teams begin estimating work for which they are responsible, they may have to 'back into' the estimates, starting at the overall budget or scope and using the number of the provided hours or dollars that have been allocated to their part of the work.



The entire team may be asked to think creatively when planning and executing the work in order to meet the overall project requirements. The top-down approach is best used when similar work has been done before. Additionally, top-down estimates are more accurate if those preparing the estimates have the appropriate level of expertise for which they are estimating.

The main disadvantage of using the top-down approach is that lower-level categories may be inaccurately estimated, and often underestimated by upper-management due to their bias on attempting to keep costs or time to completion unrealistically down.

The wedding example

An example of the top-down approach would be the instance of a bride who is enlisting the help of a wedding planner (as her *Project Manager*). The bride has limited funds (USD10,000) and has constraints as to when she wants the wedding to take place (the month of May). The wedding planner must determine the various options that can be pursued based on the available funds, as well as the time constraints indicated (the number of weekends in May).

Bottom-up approach

On the other side of the spectrum is the bottom-up approach (also referred to as expert judgement). With this strategy, those responsible for executing or actually performing the work are charged with determining the appropriate estimates.

Individuals (or team leaders) start with the WBS, assigning time estimates to each task and then using these estimates to determine the associated required completion times and the budget. Some of the advantages of using the bottom-up approach are:

- The individuals who are performing the work are better equipped to provide estimates than those who are far removed from the project effort (that is, upper management).
- Buy-in to the project budget and schedule. Because the team plays an integral role in developing the time estimates and budget, they will be more committed to adhering to their 'promise', so to speak.

The wedding example

Consider the example of the bride and wedding planner again. If the bride were to use the bottom-up approach, she would simply go to the wedding planner, indicate some of her requirements and let the planner run with her ideas (regardless of time constraints or costs).

As you can see with the scenario of the bride and wedding planner, many managers are reluctant to deploy a pure bottom-up approach for fear that the teams will inflate time and cost estimates in order to:

- ensure total success (and perhaps beyond) with their portion of the project; and
- ensure that there are sufficient funds left over to accomplish work other than that required for this project, for example, 'padding' the estimate for various reasons.

If everyone who is estimating does this, the overall project budget could become inflated to the point that the project will not be funded at all for fear of too high an expense or time of completion in doing so.



Therefore, many organisations use a combination of both strategies. The upper management may ask teams to develop bottom-up estimates and later challenge those estimates by imposing a 'target' budget.

Factors Affecting Time and Cost Estimates

Now that we have seen how organisational structure can influence a project's budget, let us look at the various techniques that can be applied to determine time and cost estimates. As you may have already experienced in your line of work, predicting an accurate time estimate can be very difficult.

The difficulty in predicting estimates stems from all types of variability such as:

- level of expertise,
- familiarity with technology,
- familiarity with process,
- familiarity with products used (possibly for the first time),
- unpredictable circumstances (for example, key staff members going on medical leave),
- resource productivity,
- type of resources used (skill level and mix) on projects, and
- availability and type of materials used.

There are others variances, to be sure. However, before we discuss the details of the techniques that can be used to effectively develop time estimates, let us try the following exercise to see just how difficult estimating can be.

Exercise: Estimating Time

The purpose of this activity is to demonstrate the difficulties the teams face when developing estimates.

This exercise presents you with a 12-piece jigsaw puzzle and challenges you to estimate the time you would need to put the puzzle together.

After trying it the first time, see what the variance is between the time you estimated and the time you actually took, then, re-estimate the time you would need to do it again.

Variances demonstrate that team members often 'shoot from the hip', so to speak, when estimating. Additionally, the level of expertise also affects one's estimates.

As you become more familiar with the puzzle and based on the knowledge of how long you took at your first attempt, your estimates should become more accurate. This improvement in estimation emphasises how working on similar activities and becoming an 'expert' in a particular field can improve estimating accuracy (expert judgment).

Estimation Techniques

Now let us briefly discuss two of the most popular estimating techniques practised in the industry:

1. Program Evaluation and Review Technique (PERT)
2. Critical Path Method (CPM)



Project Management

The two techniques are quite similar in nature. Note that PERT is rarely used in activity sequencing today; although, it is used extensively in assessing risk in a project.

According to the Project Management Institute's definition, PERT uses a weighted average to estimate expected activity duration (completion time). To calculate the estimate, PERT takes into consideration three estimates:

1. Optimistic
2. Most Likely
3. Pessimistic

PERT deals with uncertainty. There are three time estimates provided for each activity because we are not sure exactly how long it will take us to complete it. CPM does not deal with uncertainty, so there is only one time estimate provided for each activity, that is, we presume to know how long it will take to complete each activity in this implementation (CPM).

Thus, with CPM, there is no formula for computing a weighted average for the duration of an activity as there is with PERT (PERT Weighted Average), as the estimated duration is a single number.

We will discuss CPM and PERT in greater detail in the next segment.

The Budget Estimate

The development of the project budget takes into consideration:

- the WBS,
- activity duration time estimates, and
- resource assignments.

Once all project activities for each WBS element have been identified and the time estimates (durations) have been made for each activity, it is possible to begin developing the project budget.

Resource requirements need to be evaluated for each activity in the project schedule in order to derive the budget estimate. The following questions need to be addressed:

- *How* many resources will be needed?
- *What* is the cost of each resource?
- *When* will each resource be required?

It is important to note that not all resources are people. Resources can also be material (for example, sheet metal required to build the body of a new piece of equipment) or equipment (for example, powder-coating machine used to paint and plate the required components in a project for developing a new piece of equipment).

While making the budget estimate, you will have to calculate the costs that contribute to project expenses and the ensuing project budget. Table 3 lists different cost categories:

Table 3: Cost Categories

Cost	Example
Labour cost	Resources



Project Management

Material cost	Sheet metal
Equipment cost	Use of a power-coating machine
Direct costs	Travel, phone, contracted services
Indirect costs	Operating expenses such as a company's overhead

Whether top-down or bottom-up estimating (or a combination of these two) is applied, the process of calculating project costs is the same. Once *labour, material, equipment, and other direct and indirect* costs have been considered, the cost for each activity is calculated considering the time estimate provided for it.

Wedding Example:

Consider the example of the bride and wedding planner once again. Let us say that the bride provides the following list of items (or activities) as requirements for her wedding:

- a string quartet to play before, during and after the wedding ceremony;
- a church for four hours to hold approximately 150 guests;
- a soloist to sing during the ceremony; and
- two limousines to transport the wedding party from the church to the reception hall.

The wedding planner sets out to find the necessary resources to complete each of the activities listed above. The wedding planner then multiplies the time estimate for each activity against the *per-unit* resource dollar rate (when required) in order to estimate the total cost of completing the activity.

Requirement	Rate	Duration	Expense
A string quartet to play before, during and after the ceremony	USD 250/hour	3 hours	USD 750 (labour)
A church for 4 hours to hold approximately 150 guests	Flat rate	4 hours	USD 1,000 (Contracted expense for facility rental)
A soloist to sing during the ceremony	Flat rate		USD 400 (labour)
Two limousines to transport the wedding party from the church to the reception hall	USD 200/limousine per hour	2 hours	USD 800 (Contracted expense for equipment rental)



Project related costs	USD 2,950
Other direct and indirect costs (35% of USD 2,950)	USD 1,032
Total costs for these activities	USD 3,982

Similar to the way the wedding planner developed her budget for this portion of the project, a *Project Manager* will consider the costs of each project activity, and, where appropriate to do so, will multiply the usage rate against the estimated time and add these to determine the overall project budget.

Summary

In this segment, you learned that:

- once the following have been determined, the cost for each activity can be calculated against the time estimate provided for it:
 - labour
 - material
 - equipment
 - other direct, and
 - indirect costs;
- many organisations use a combination of both a top-down approach and a bottom-up approach to estimate time and budget for a project; and
- time or activity duration is estimated using PERT (when we have uncertainty about the time required to complete an activity) or CPM (when the time required to complete an activity is felt to be known).



PERT and CPM

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1. Objectives
2. Introduction
3. Development of Project Network
4. PERT
5. CPM
6. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the development of project network
- Explain the PERT model
- Explain the CPM model

Introduction

In the previous segment, we understood the need for, and the process of, estimating activity durations and the two fundamental network techniques, namely PERT or Program Evaluation Review Technique and CPM or Critical Path Method. In this segment, we will understand PERT and CPM in detail.

PERT was originally developed to facilitate the planning and scheduling of the Polaris Fleet Ballistic Missile Project of the US government. Designed to handle risk and uncertainty, PERT is quite suitable for research and development programmes, aerospace projects and other projects related to new technology. In such projects, there exists variability in the time requirement for completing different jobs or activities. Therefore, the orientation of PERT is 'probabilistic'.

CPM is similar to PERT. It was developed in the US by the DuPont Company in 1956-57 for solving scheduling problems of industries. CPM is principally concerned with the matters pertaining to cost and time. Its application is mainly to projects that use a fairly stable technology and are quite risk free. Therefore, its orientation is 'deterministic'.

Extensively diverse projects are open to analysis by PERT and CPM. For example, launching a spaceship, research and development programmes, constructing a plant, building a river valley project, overhauling an organisation, training manpower, starting a new venture and adult literacy programme are all examples of such projects. This segment focusses on the basics of PERT and CPM.

Development of project network

The network diagram is fundamental to understanding PERT and CPM. The network diagram, also known as project graph, depicts the project activities and events and their logical relationships. Figure 1 depicts a simplified network diagram for a dinner project.

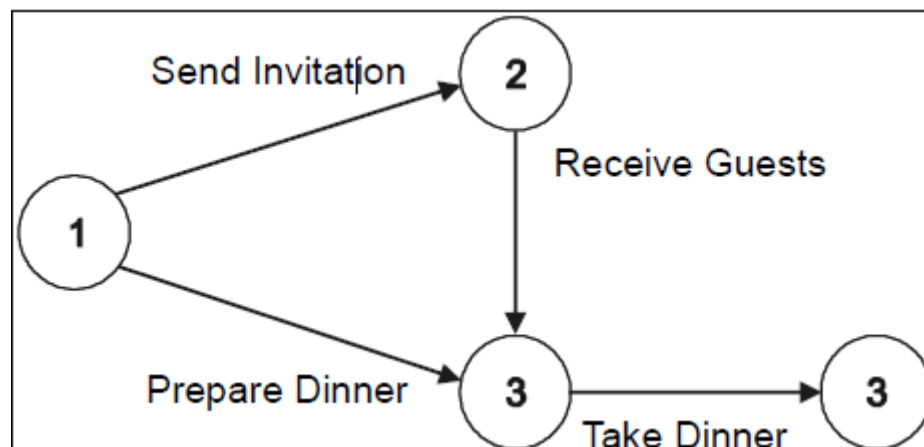


Figure 1

Activities and events are used to construct a network diagram. An activity is a specific task, job or function to be carried out in a project. For example, 'prepare dinner' (see Figure 1) is an activity. An activity is symbolised by an arrow. The head of the arrow depicts the completion of the activity and the tail of the arrow depicts its beginning. (There is no significance of length and 'compass' direction of the arrow.) The event is a specific point in time representing the beginning or end of one or more activities. It stands for a milestone and does not consume time or resources.

It is essential to detail all the activities of the project because activities are the basic building blocks of a network diagram. For this reason, it is useful to break the project into some steps. The number of steps in a project depends upon its magnitude and complexity. For industrial projects, usually, a two-step procedure is sufficient. In the primary step, the major parts of the project are identified and in the next step, the activities of each major part are defined. Activities must be so defined that they are distinct, logically uniform tasks for which time and resource requirement can be estimated.

After specification of activities, it is essential to define for each activity, the activities which precede it, the activities which succeed it and the activities which can take place simultaneously. After getting this information, the network diagram is developed through forward method or backward method, showing the logical relationship among activities and events.

The forward method starts with the initial event, showing the beginning of the project, and proceeds forward till the end event is accomplished. The backward method starts with the end event and works backwards till the beginning event is reached.

PERT

For using the PERT model, we need to consider the variability of project duration.

Measures of variability

Variability in PERT analysis is measured by variance or its square root, that is, standard deviation. Variance of a set of numbers is the average squared difference of the numbers in the set from their arithmetic average. A simple example may be given to illustrate the calculation of variance. Let us take a series that consist of numbers 4, 6 and 8. The average of this series is 6.



The differences of various numbers in the series from this average are -2, 0 and 2. Squaring them we get 4, 0 and 4. Hence, the variance—the average of squared difference—is $8/3$ and standard deviation is $\sqrt{8/3}$. The steps involved in calculating the standard deviation of the duration of critical path are as follows:

Step 1: Determine the standard deviation of the duration of each activity on the critical path.

Step 2: Determine the standard deviation of the total duration of the critical path on the basis of information obtained in step 1.

In order to determine the standard deviation of the duration of an activity, we require the complete probability distribution of the activity distribution. We, though, have just three values from this distribution: t_p , t_m , and t_o . In PERT analysis, a simplification is used for calculating standard deviation. It is calculated by the following formula:

$$\text{Standard deviation} = (t_p - t_o)/6$$

Where, t_p = pessimistic time
 t_o = optimistic time

Variance is obtained by squaring standard deviation.

The standard deviation and variance of the activities on the critical path of our illustrative project are shown in the Table 4:

Table 4: Standard Deviation and Variance of the Activities

Activity	t_p	t_o	$O = (t_p - t_o)/6$	Variance = O^2
(1-2)	21	9	2	5.00
(2.5)	24	10	2.33	5.43

Assuming that the probability distribution of various activities on the critical path is independent, the variance of the critical path duration is obtained by adding the variance of activities on the critical path.

In real life projects which have a large number of activities on the critical path, we can sensibly presume that the critical path duration is normally distributed, with mean and standard deviation attained.

A normal distribution looks like a bell shaped curve as shown in Figure 2. It is symmetric and single peaked and is fully described by its mean and standard deviation. The probability of values lying within certain ranges is as follows:

Range	Probability
Mean $\pm$ One standard deviation	0.682
Mean $\pm$ Two standard deviations	0.954
Mean $\pm$ Three standard deviations	0.998

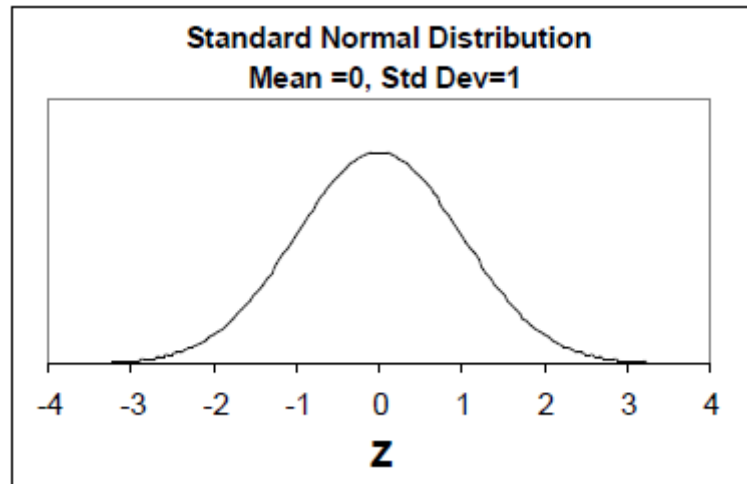


Figure 2: Normal Distribution

Probability of completion by a specified date

Armed with information about mean (T) and standard deviation (a) for critical path duration, which is normally distributed, we can compute the probability of completion by a specified date (D) as follows:

- Find $Z = (D - T)/S.D$, where Z is known as standard normal variate with mean and variance.
- Obtain cumulative probability up to Z by looking at the probability distribution of the standard normal variate.

Table 5: Cumulative Probability up to Z for Standard Normal Distribution

Z	Cumulative Probability
-3.0	0.001
-2.8	0.003
-2.6	0.005
-2.4	0.008
-2.2	0.014
-2.0	0.023
-1.8	0.036
-1.6	0.055
-1.4	0.081
-1.2	0.115
-1.0	0.159
-0.8	0.212
-0.6	0.274
-0.4	0.345
-0.2	0.421
0.0	0.500
0.2	0.579
0.4	0.655



0.6	0.726
0.8	0.788
1.0	0.841
1.2	0.885
1.4	0.919
1.6	0.945
1.8	0.964
2.0	0.977
2.2	0.986
2.4	0.992
2.6	0.995
2.8	0.997
3.0	0.999

The above procedure may be illustrated for our project which has $T = 28$ and $SD = 3.07$. Table 3 depicts the probability of completing this project by certain specified dates.

Table 6: Probability of Completing the Project

Specified	Z	Probability of Completion by D
20	$\frac{20-28}{3.07} = -2.6$	0.005
25	$\frac{25-28}{3.07} = -1.0$	0.159
30	$\frac{30-28}{3.07} = 0.6$	0.726

CPM

The PERT model was developed for projects, which are considered uncertain; the CPM model was developed for projects, which are comparatively risk-free. Both the approaches start with the development of the network and a focal point on the critical path. The PERT approach is 'probabilistic' while the CPM approach is 'deterministic'.

This does not, however, mean that in CPM analysis we work with single time estimates. Actually the main focus of CPM analysis is on variations in activity times as a consequence of changes in resource assignments.

These variations are planned plus related to resource assignments and are not caused by random factors outside the control of management as in the case of PERT analysis. The major focus of CPM analysis is on time-cost relationships and it seeks a project schedule that minimises total cost.

Assumptions

The usual assumptions underlying CPM analysis are:

1. The costs associated with a project can be divided into two components: direct costs and indirect costs. Direct costs are incurred on direct material and direct labour. Indirect costs consist of overhead items like indirect supplies, rent, insurance, managerial services, etc.



Project Management

2. Activities of the project can be expedited by crashing which involves employing more resources.
3. Crashing reduces time but enhances direct costs because of factors such as overtime payments, extra payments and wastage. The relationship between time and direct activity cost can be reasonably approximated by a downward sloping straight line. Figure 5.12 depicts a typical cost-time line.

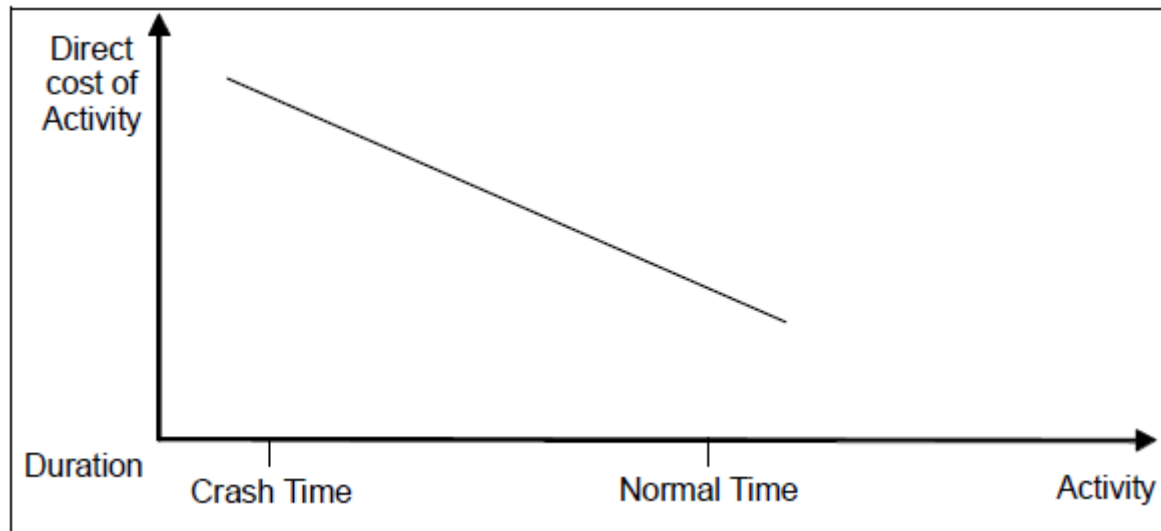


Figure 3: A typical cost-time line

4. Indirect costs associated with the project increase linearly with project duration. Figure 4 depicts a typical line for indirect costs.

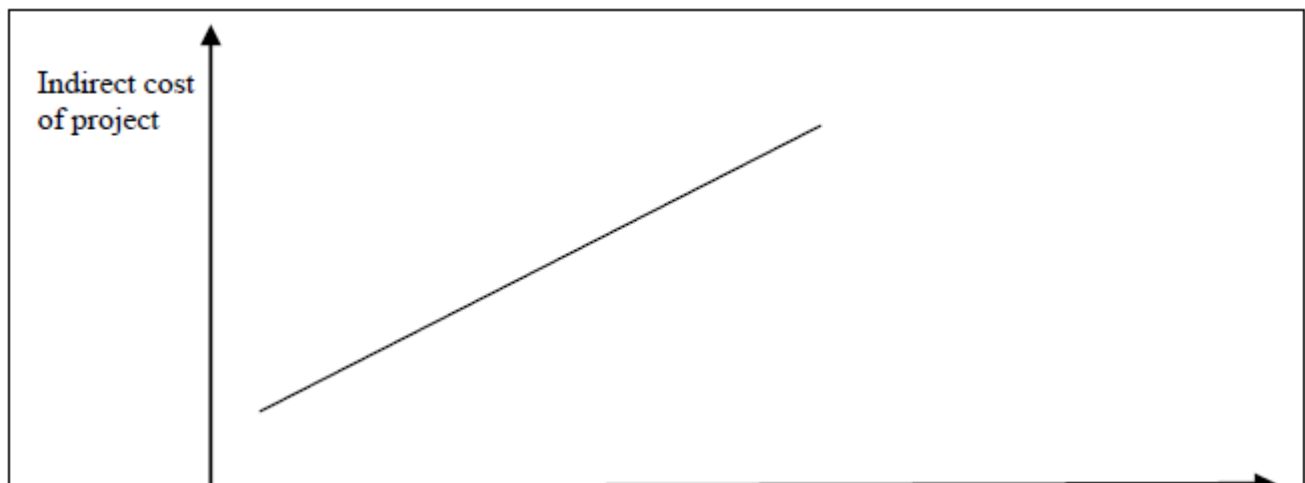


Figure 4: Indirect costs

Procedure

Given the above assumptions, the CPM analysis seeks to examine the consequences of crashing on total cost (direct cost plus indirect cost). Since the behaviour of indirect project cost is well defined, the bulk of CPM analysis is concerned with the relationship between total direct cost and project duration. The procedure used in this respect is generally as follows:



Step 1: Obtain the critical path in the normal network. Determine the project duration and direct cost.

Step 2: Examine the cost-time slope of activities on the critical path obtained and crash the activity, which has the least slope.

Step 3: Construct the new critical path after crashing as per step 2. Determine project duration and cost.

Step 4: Repeat steps 2 and 3 till activities on the critical path (which may change every time) are crashed.

Example: The above procedure may be illustrated with an example. Table 7 depicts the activities, durations and direct activity costs of a project. The indirect cost is Rs. 2,000 per week.

Table 7: Normal and Crash Time and Cost

Activity	Time in Weeks		Cost		Cost to Expedite per Weeks
	Normal	Crash	Normal	Crash	
1-2	8	4	3,000	6,000	450
1-3	5	3	4,000	8,000	2000
2-4	9	6	4,000	5,500	500
3-5	7	5	2,000	3,200	600
2-5	5	1	8,000	12,000	1,000
4-6	3	2 ½	10,000	11,200	2,400
5-6	6	2	4,000	6,800	700
6-7	10	7	6,000	8,700	900
5-7	9	5	<u>4,200</u> 45,200	<u>9,000</u> 70,400	1,200

Figure 5 depicts the project network with normal duration.

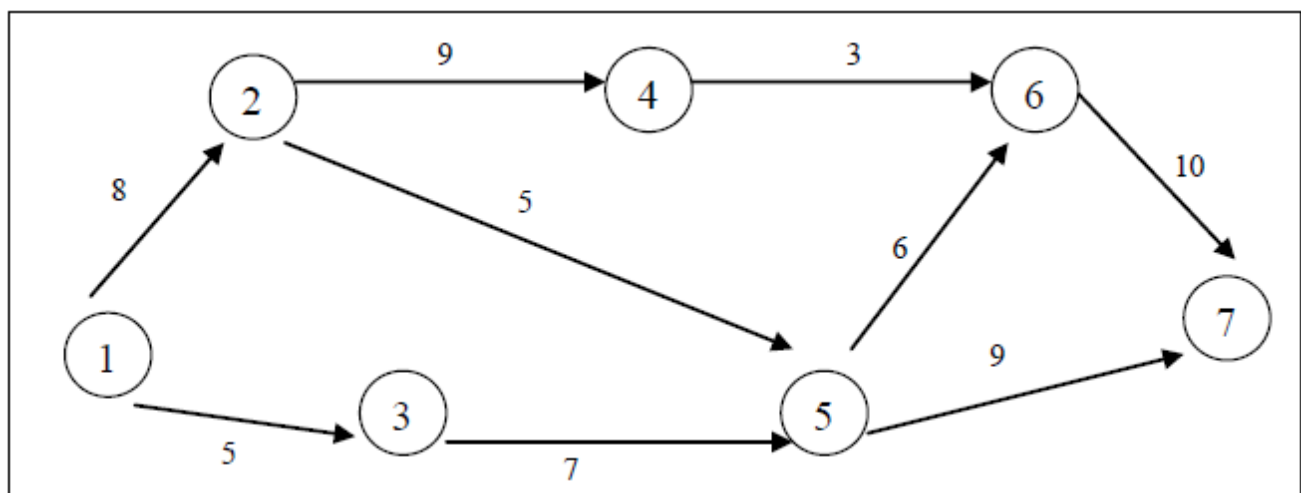




Figure 5: Project Network (1)

The critical path in the all normal network is (1-2-4-6-7). The project duration is 30 weeks and the total direct cost is Rs. 45,200. Examining the time cost slope of activities on the critical path, we find that activity (2-4) has the lowest slope; in other words, the cost to expedite per week is the lowest for activity (2-4). Hence, activity (2-4) is crashed. Figure 6 depicts the project network after such a crashing.

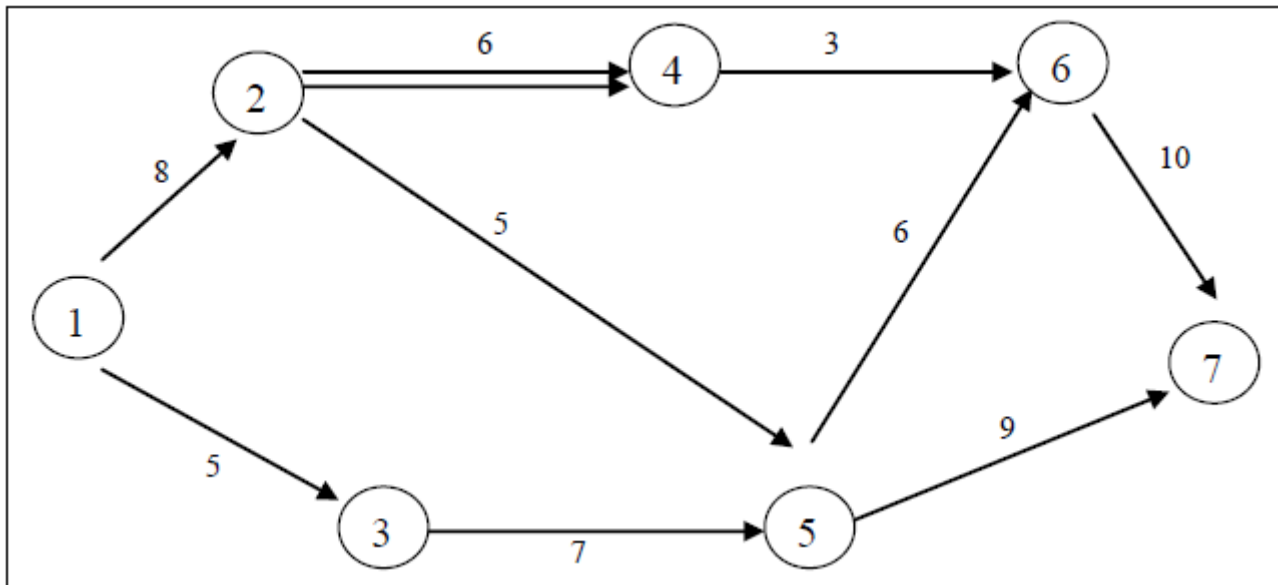


Figure 6: Project Network (2)

As per Figure 6, the critical path is (1-2-5-6-7), with a length of 29 weeks, and the total direct cost is Rs. 46,700. Looking at the time-cost slope of the activities on the new critical path (1-2-5-6-7), we find that the activity (5-6) has the lowest slope. Hence, this activity is crashed. Figure 6 depicts the project network after such crashing, the critical path is (1-2-4-6-7) with a length of 27 weeks and the total direct cost is Rs. 49,500.

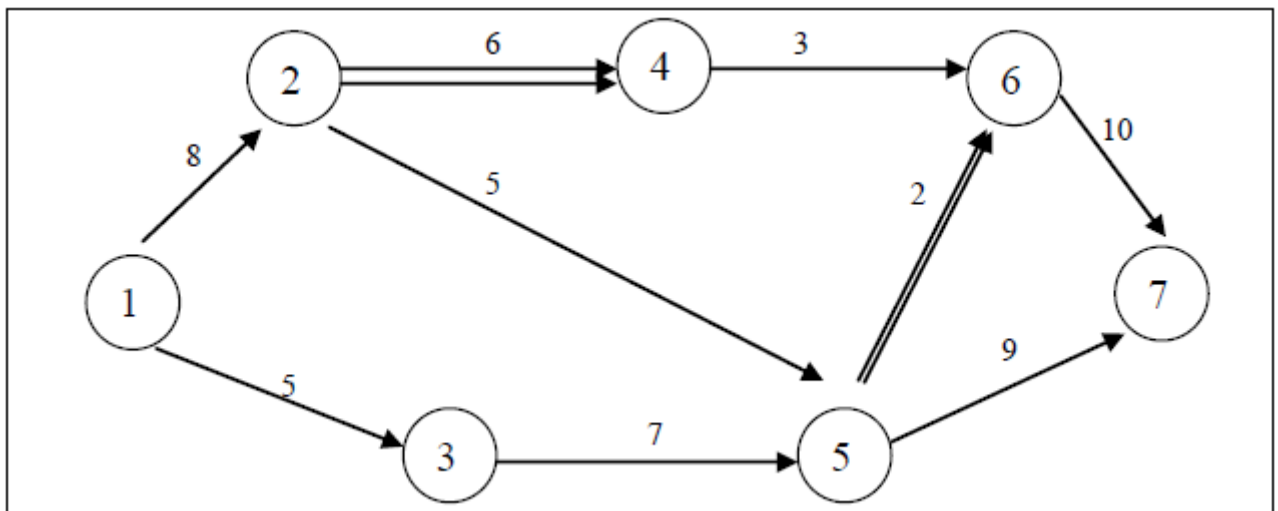




Figure 7: Project Network (3)

Comparing the time-cost slope of the non-crashed activities on the new critical path (1-2-4-6-7), we find that the activity which costs the least to crash is (1-2). Hence, this is crashed. Figure 8 depicts the project network after such a crashing. As per Figure 8, the critical path is (1-3-5-6-7) with a length of 24 weeks and the total direct cost is Rs. 52,500.

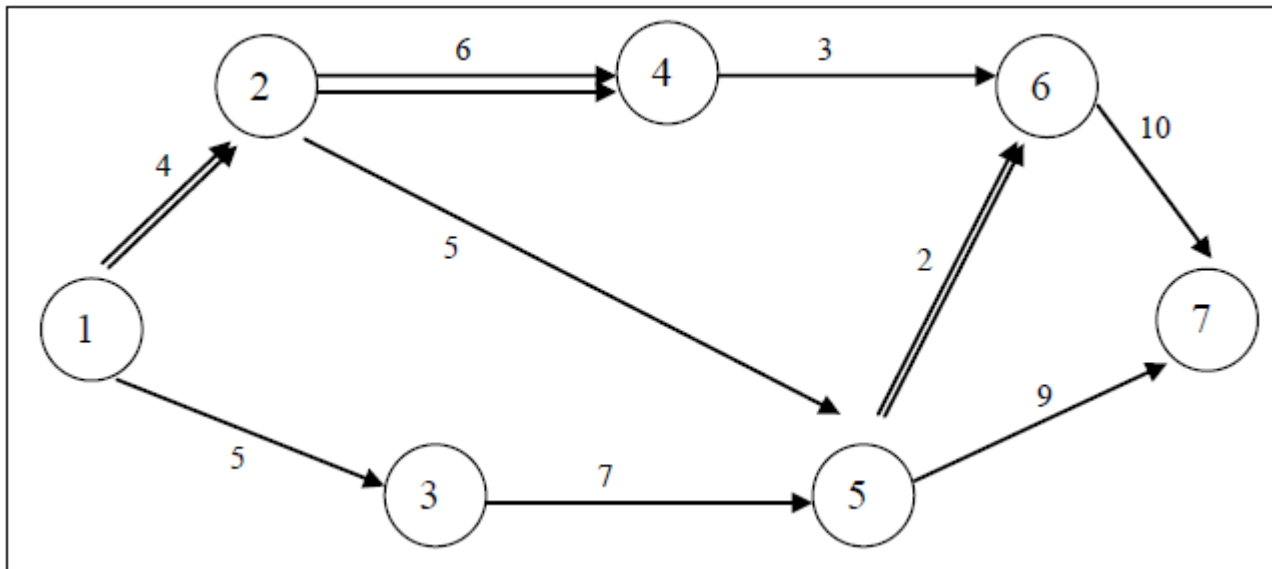


Figure 8: Project Network (4)

Looking at the time cost slope of the non-crashed activities on the new critical path, (1-3-6-7), we find that activity (6-7) has the lowest slope. Hence, it is crashed. Figure 9 depicts the project network after such a crashing. As per Figure 9, there are two critical paths (1-3-5-6-7) and (1-3-5-7), both with a length of 21 weeks, and the total direct cost is Rs. 55,200.

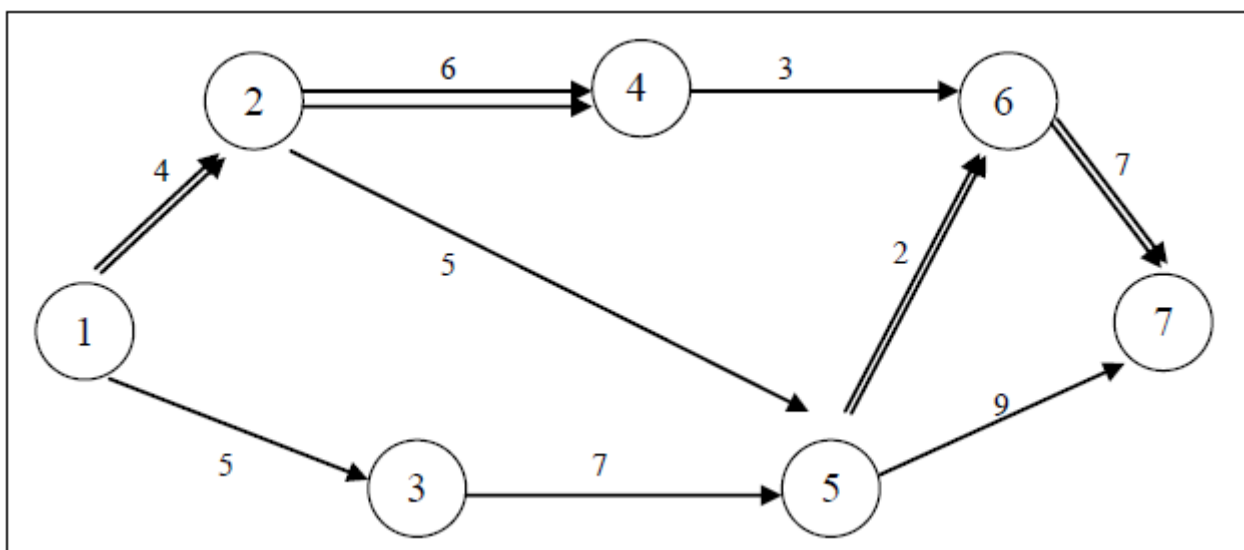




Figure 9: Project Network (5)

Considering the time-cost slope of non-crashed activities on critical paths (1-3-5-6-7) and (1-3-5-7), we find that activity (3-5), which is common to both the critical paths is the least costly to crash. Hence, it is crashed. Figure 10 depicts the project network after this crashing. As per Figure 10, the critical path is (1-2-4-6-7) with a duration of $20\frac{1}{4}$ weeks and the total direct cost is Rs. 56,400.

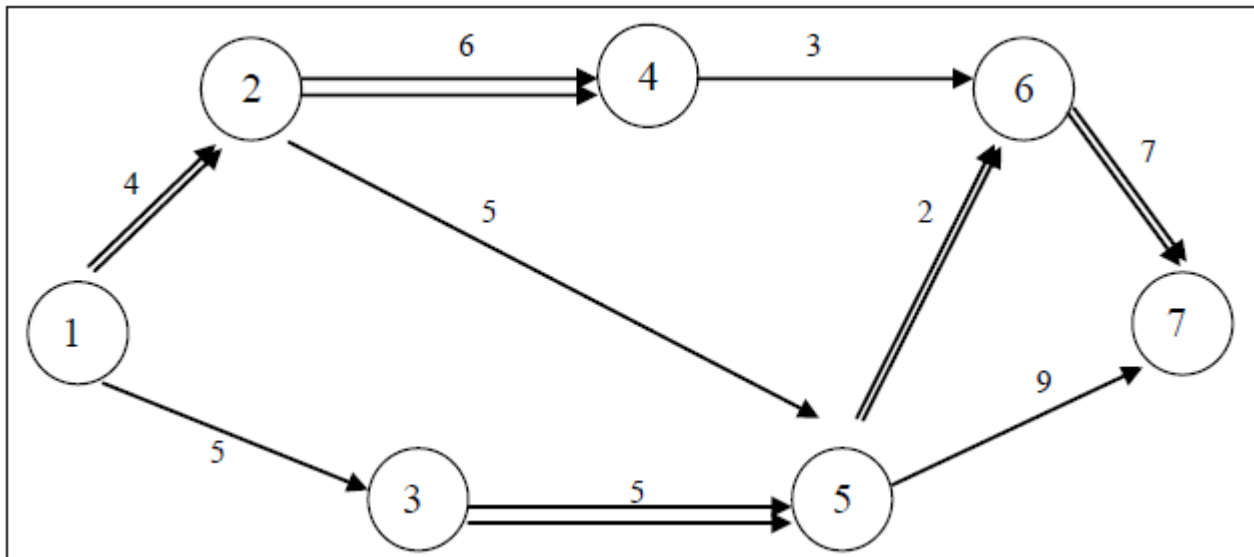


Figure 10: Project Network (6)

Looking at the new critical path (1-2-4-6-7), we find that the only non-crashed activity is (4-6). Figure 11 depicts the project network given by crashing this. As per Figure 11, the critical path again is (1-2-4-6-7) with a duration of $19\frac{1}{2}$ weeks and the total direct cost is Rs. 57,600.

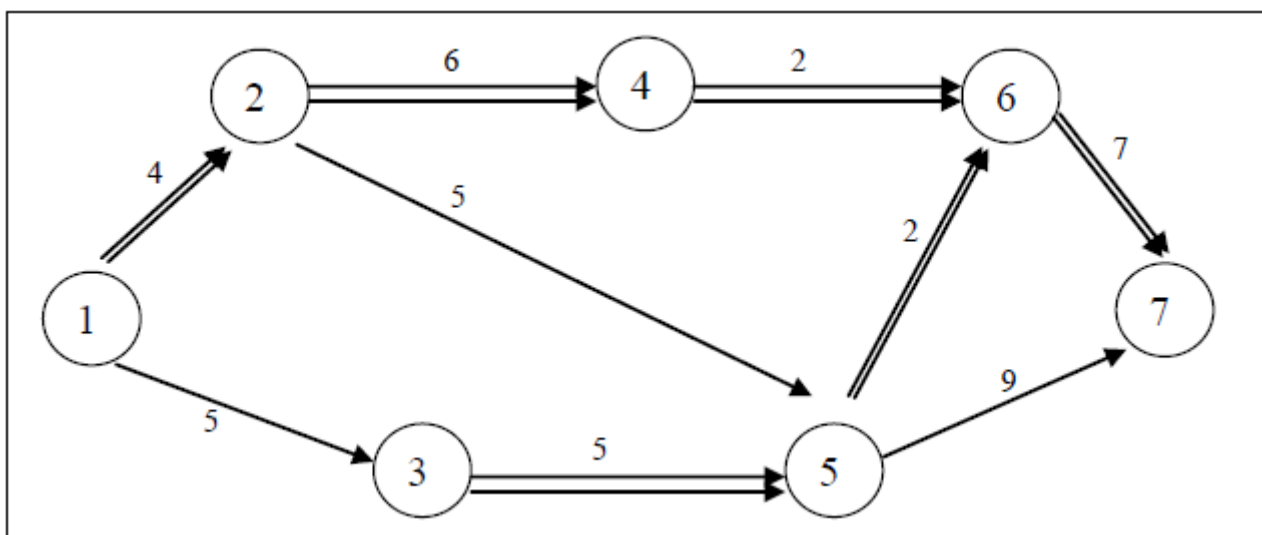


Figure 11: Project Network (7)

Since all the activities on the critical path (1-2-4-6-7) are crashed, there is no possibility of further time reduction. Hence, let us now look at the time-cost relationship. This is shown in Figure 11.



From table 8, we find that the total cost is minimised for the project schedule represented by the activities crashed, which are (1-2), (2-4), (3-5), (5-6), (6-7). The information provided in table 8 is useful for decision-making. Table 8 depicts the project duration and total cost.

Table 8: Project duration and total cost

Figure no.	Activities Crashed	Project Duration in Weeks	Total Direct Cost	Total Indirect Cost	Total Cost
5.14	None	30	45,200	60,000	105,200
5.15	(2-4)	29	46,700	58,000	104,700
5.16	(2-4) and (5-6)	27	49,500	54,000	103,500
5.17	(1-2), (2-4) and (5-6)	24	52,500	48,000	100,500
5.18	(1-2), (2-4), (3-5), (5-6) and (6-7)	21	55,500	42,000	97,200
5.19	(1-2), (2-4), (3-5), (3-5) and (6-7)	20	56,400	40,000	96,400
	(1-2), (2-4), (3-5), (5-6) and (6-7)	19 ½	57,600	39,000	96,600

If the objective is to minimise the total cost of the project, the pattern of crashing suggested by Figure 10 is optimal. If the objective is to minimise the project duration, then the pattern of crashing suggested by Figure 11 is optimal. In real-life situations, however, both the factors may be important. In addition, factors such as strain on resources and degree of manageability are also important. The final decision would involve a careful weighing and balancing of these diverse factors, some quantitative and some qualitative.

Summary

In this segment, you learned that:

- PERT and CPM are the two fundamental network techniques,
- the orientation of PERT is probabilistic whereas that of CPM is deterministic,
- PERT is suited to projects that are considered uncertain,
- CPM is suited to projects that are comparatively risk-free, and
- Both, PERT and CPM, start with the development of the network and a focal point on the critical path.



Identifying Critical Activities

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Objectives

Upon completion of this segment, you will be able to:

- Compute critical path statistics
- Identify the critical activities in a project
- Describe the term 'float'
- Explain how network diagrams can help identify the critical path and related activities of the critical path.

Introduction

In the previous segment, we learned about PERT and CPM in detail. In this segment, we will focus on identifying the critical activities in a project.

It is possible to sequence project activities using precedence diagramming and arrow diagramming methods, and display activity relationships using graphical depictions such as Gantt charts and Program Evaluation Review Technique (PERT) charts. In this segment, we will combine these methods and charting procedures to determine a project's critical path and compute the critical path statistics for a project.

Critical Path

According to the Project Management Institute (PMI), critical path is defined as 'the series of activities that determine the duration of a project.' In a deterministic model, the critical path is usually defined as 'those activities that possess float which is less than or equal to a specified value, often zero. It is the longest path through the project.' In other words, when the durations of each of the activities on this path are added, or summed, it is the longest path through the project or the one with the maximum possible duration.

Float can be described as the amount of time an activity can be delayed, from its earliest start time, before its delay causes a delay in the entire project. It is important to recognise that a delay in any of the activities on the critical path will cause an overall delay in the project beyond the earliest possible completion time.

Network diagrams

Network diagrams and charts can help determine a project's critical path length as well as identify which activities are on the *critical path*. Network diagrams are also useful for identifying activities



Project Management

that have abundant *float*. Identifying these types of activities can be useful in the project when the delay of an activity on the critical path occurs.

It might be possible to reassign resources to the activities without float (time-critical activities) to bring the project back on track for a successful timely completion. This type of technique is often known as *project crashing* or *resource reallocation*.

Now, let us see a demonstration of network diagrams.

When using network diagrams to determine a project's critical path length, you must consider the *start* and *finish* times for each activity.

Both start and finish should be considered in terms of early start (ES)/late start (LS) and early finish (EF)/late finish (LF) times. These four quantities plus activity float comprise the *critical path statistics* of a project.

When discussing critical path calculations for this subject, please note the following assumptions:

- We will use the earliest start and finish dates to calculate the critical path. Using this assumption, all activities on the critical path will have zero float.
- It will be assumed that our projects will start on time and that no activities will have a negative float, causing us to plan them to be late from the start.

Typically, the critical path calculation for a project will analyse every project activity. For our demonstration purposes, we may only use a subset of the project activities to simplify the manual calculations. Scheduling software such as P3E/c can easily calculate the critical path for a project containing several thousands of activities, with a click of a button.

The *early start* of an activity is the earliest time period or date that the activity can begin after all of its predecessor activities have finished. The *early finish* of an activity is the earliest possible point in time when the activity can conclude. It can be calculated by considering the activity's early start time and adding its estimated completion time (duration).

The *late start* of an activity is the latest possible point in time the activity can begin without delaying the overall schedule beyond the critical path length. Likewise, the *late finish* of an activity is the latest possible point in time an activity can finish without delaying the overall project schedule.

Activity A has an early start (ES) of 0 and a late start (LS) of 2. Its early finish (EF) is 2 while its late finish (LF) is 4. Because there is a precedence relationship between activity A and B (Activity A must finish before Activity B can start), activity B's early and late *start* are dependent upon activity A's early and late *finish*.

Because the early and late starts (and early and late finishes) for these activities is different, there is slack present in scheduling them. For example, Activity A can start immediately (ES = 0), but need not absolutely start for 2 periods (LS = 2) and the overall completion period of the project (LF of Activity 'Finish') will not be affected.

Thus, we say that Activity A has two units of slack in it ($LF - EF = 4 - 2 = 2$).

Considering the network logic, the critical path consists of activities:

Start->C->D->E->F->G->Finish



This is the only other path in this network, and we have already established activities on the first path we identified or the upper path, which have positive amounts of slack in them (both Activities A and B). If Activities A and B are not on the critical path by virtue of possessing positive amounts of slack or total float, then Activities C, D, and E must be the remaining activities in this network required to make a continuous path through it, that is, activities Start, F, G, and Finish.

Summary

In this segment, you learned that:

- The longest path through a network, or the critical path, gives the minimum time in which the project can be completed. It might require more time to complete in actuality, but it cannot require any less.
- Network diagrams can be useful in determining a project's critical path length, as well as to identify which activities are on the critical path.



Project Quality Management and Value Engineering

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1. Objectives
2. Introduction
3. Quality
4. Concepts of Quality
5. Value Engineering
6. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the meaning and importance of quality
- Evaluate the quality concepts
- State the implication of quality planning, quality assurance, and quality control for project quality management
- Define Value Engineering (VE)

Introduction

In the previous segment, we learned how to determine a project's critical path and compute the critical path statistics for a project. In this segment, we will understand the concepts of quality and value engineering.

Good projects ensure quality processes for a standard output. Quality management principles are used in conjunction with the project management tools to ensure that the project deliverables are in tune with customers' expectations. It is important to monitor the project activities continuously to maintain quality standards and avoid mishaps.

'Quality' refers to the ability of a process or product to satisfy both stated and implied goals as defined by the stakeholders.

Quality has become a key economic factor in terms of staying competitive on a global basis. The quality management of a project means that the project should meet the needs that it was originally meant to meet.

This segment also covers the concept of value engineering. Value engineering (VE) is an effective productivity improvement tool. It originated during World War II in General Electric Company (GEC). Over the years, its usage spread to the whole world. Now, it is being practised almost in every sector of the economy in different countries in varying degrees. However, its full potential is yet to be realised by organisations.

Quality

Quality begins with designing a product as per customer specifications. Further, it covers the application of standard measuring units, the suitable raw material, selecting appropriate



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manufacturing process, etc. Quality is a relative term and it can be better understood by a comparison with the utility of the product.

Crosby defined it as, 'Quality is conformance to requirement or specifications'. Juran defined it as 'Quality is fitness for use'. The Quality of a product or service is the fitness of that product or service for meeting or exceeding its intended use as required by the customer.

Characteristics of quality

Quality characteristics such as performance, functionality, suitability, reliability and consistency are commonly used to measure project deliverables. These quality characteristics are not only applicable to the deliverable itself, but also to the processes or methods used to deliver the final product, such as equipment or computers. Therefore, the required quality characteristics must be fully stated and understood before procurement of any services, materials or equipment used in the project.

Companies that offer a powerful combination of low prices and high quality are capturing the hearts and wallets of consumers. Today, as companies are becoming more value driven, there is a shift in competitive terrain from 'price orientation' to 'quality and convenience' orientation.

As these value-driven companies gain share, at varying speeds, across economies, they change the nature of competition by transforming consumer attitudes about trade-offs between price and quality. Today, quality is considered an 'order qualifying' rather than an 'order winning' attribute of the product or service.

So, how do consumers assess quality?

One approach that customers use to evaluate quality is to mention attributes of the product or its product delivery process. For example, if someone were to ask you to assess the quality of a notebook computer, you might reply by referring to things such as: the way it looks, aesthetic appeal, system configuration—how long it took to set up, hard disk capacity and other software installed—how long it takes to boot up and whether or not it has an Intel Chip Inside.

In the case of services, such as a meal at a restaurant, the determinants of quality might include: the meal itself, its presentation, aroma of the food, the manner in which it was served, behaviour of the waiter, behaviour of the other diners at the neighbouring tables and the overall ambience of the place.

In effect, you are citing attributes of quality, that is, the traits associated with quality that can be identified and, more importantly, measured. However, attributes are not the same as quality. Identifying every attribute of quality for a product would not describe that product's quality level.

Some attributes used to help define quality are:

Freshness

Some products are perishable, that is, their quality declines over time. For example, vegetables fall under this category. Fashion items also are subject to obsolescence. At the other extreme, the value associated with some products increases with age, as is the case with antiques and red wine.



Reliability

The quality associated with a product often increases with the dependability of the product and customer experience. For example, the patients expect the hospitals to have competent staff, and the customers expect telephones to work.

Durability

The quality attribute that implies product performance under adverse conditions. For example, Timberland's commercials are designed to convey the durability of its outdoor footwear.

Safety

This is an attribute of quality that measures the likelihood of harm from goods or service. What is safe can be a controversial issue. For instance: is a gun with a safety clip safe? Is the packaging of a product tamper proof?

Environmental friendliness

As is the case with safety, this quality attribute has both societal aspects and is individual specific. The requirements for being considered an environmentally friendly product are becoming more stringent. For example, firms must now also focus on how a product is disposed of after its useful life.

Serviceability

This attribute relates to the ease and cost associated with servicing a product after the sale has been made. Products are now being increasingly designed so that they do not need service; for example, car batteries. But many others do require service and this capability must be both designed into the product and the post-sale service system. This is especially important for consumer durables.

Attribute consistency

The attributes associated with a product should be internally consistent. It would make little sense to build a car with air-foils or a biodegradable cigarette filter. Products with inconsistent combinations of features are not likely to match the needs of their buyers.

To a certain degree, the functionality and the quality overlap. Products with excellent designs will excel in attributes that matter. These, in turn, increase the functionality of the product.

From the above meaning and definitions of quality, we can conclude that quality plays an important role in the overall project performance. The absence of quality will result in:

- absence of any standards for comparing the quality of goods/services,
- customer dissatisfaction due to increased maintenance and operating costs of products/services,
- increased rework cost while manufacturing products/providing services,
- reduced life time of the products/services, and
- reduced flexibility with respect to usage of standard spare parts.



Concepts of Quality

New dimensions to the concept of quality have emerged in addition to the folklore philosophy. Conventional definitions are aimed at conformance to standards while the modern thinkers focus on deriving 'value for money'.

Quality can be referred to as a state in which value entitlement is realised for the customer and supplier in every aspect of the business relationship. 'Value' represents the economic worth, practical utility and availability for customer and the company that creates the product or service. This definition acknowledges the fact that the quality of products or services rarely comprises a single element.

'Value entitlement' means:

For the customer: a fair level of expectation to buy high-quality products at the lowest possible cost.

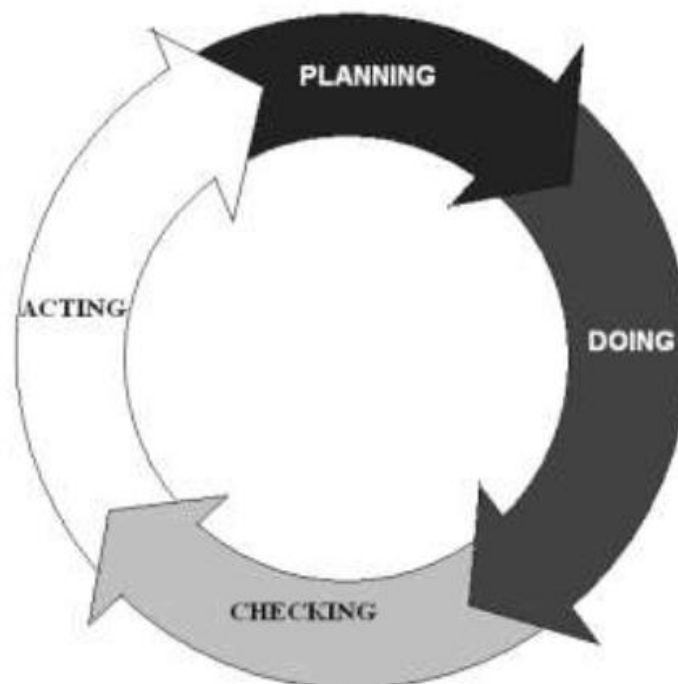
For the provider: a fair level of expectation to create quality products at the highest possible profits.

The philosophical leaders of the quality movement, notably Phillip Crosby, W. Edward Deming and Joseph S. Juran provide different perceptions to the concept of quality. It is because they use different frameworks for defining quality.

Let us understand an important management method put forward by W. Edward Deming. It is known as the Shewhart Cycle.

The Shewhart Cycle

PDCA (plan–do–check–act) is an iterative four-step management method used by companies to control and continually improve their processes and products. PDCA is also referred to as the Deming circle/cycle/wheel, Shewhart cycle, control circle/cycle, or plan–do– study–act (PDSA).





Let us now discuss what each of these steps states.

Plan

Establish the objectives and processes essential to deliver results in agreement with the expected output (the target or goals). With the establishment of output expectations, the completeness and correctness of the specification also becomes a part of the targeted improvement. When possible, start on a small scale to test likely effects.

Do

Execute the plan, implement the process, and make the product. Collect data to be used in the charting and analysis in the following 'Check' and 'Act' steps.

Check

Study the results achieved (measured and collected in the above step) and compare against the expected results (targets or goals from the 'Plan') to find out any mismatch between the two. Look for deviation in implementation from the plan and also look for the appropriateness/completeness of the plan to allow the execution, that is, 'Do'. Charting data can make it much easier to see trends over several PDCA cycles and to convert the collected data into meaningful information, which is what we need for the next step 'Act'.

Act

Request corrective action on considerable differences between actual and expected results. Analyse the differences to find out their root cause. Determine the areas where changes could be applied to improve the process or product. If completion of these four steps does not show any need for improvement, then the scope to which PDCA is applied may be fine-tuned to plan and improve with more detail in the subsequent iteration of the cycle, or attention needs to be put in a different stage of the process.

Project quality management

Project quality management begins by defining the quality standards to be used for the project. This definition will come from the stakeholders, beneficiaries and often from the overall standards for the organisation. Careful identification of the quality standards will help to ensure a successful project outcome that will be accepted by the stakeholders.

In addition to quality standards for the end result of the project, there may also be organisational quality standards that must be met for the actual management of the project, such as certain types of reporting or project tracking methods.

It describes the processes required to ensure that the project will satisfy the needs for which it was undertaken. The knowledge area of project quality management includes the organisational processes that determine the quality policies, objectives and responsibilities. It consists of quality planning, quality assurance and quality control. Figure 12 depicts the project quality management process.

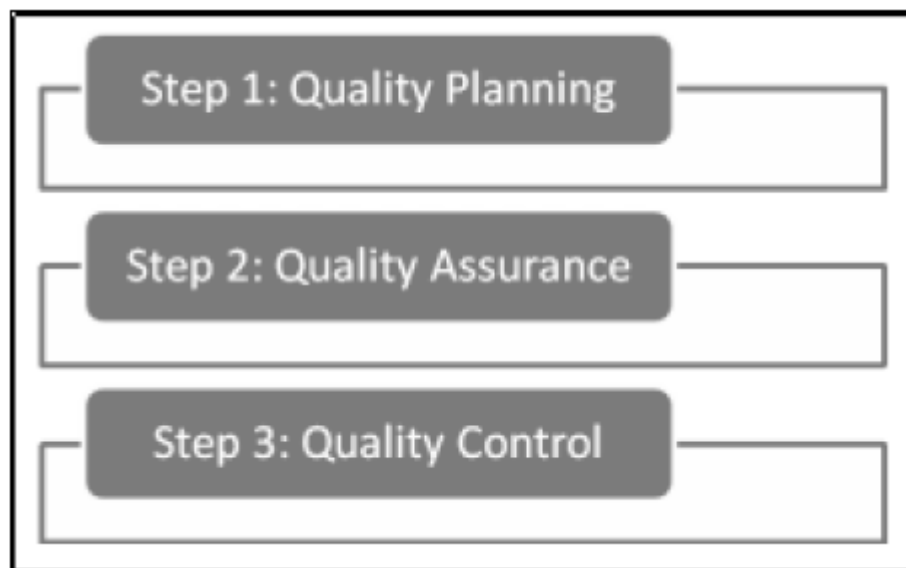


Figure 12: The Project Quality Management Process

Table 9 depicts an overview of project quality management.

Table 9: Project Quality Management

	Quality Planning	Quality Assurance	Quality Control
Input	Quality policy Scope statement Product description Standards and regulations	Quality management plan Results of quality control Measurements Operational definitions	Work results Quality management plan Operational definitions Checklists
Tools and Techniques	Benefit/cost analysis Benchmarking Flow-charting Design of experiments	Quality planning tools and techniques Quality audits	Inspection Control charts Pareto diagrams Statistical sampling Flow-charting Trend analysis
Output	Quality management plan Operational definitions Checklists Inputs to other processes	Quality improvement	Quality improvement Acceptance decisions Rework Completed checklists Process adjustments



Let us now discuss the project quality management process in detail.

Step 1: Quality planning

Quality planning is the process of identifying the quality standards that are related to the project and determining how these standards can be achieved. It is one of the significant processes of project planning and should be performed on a continuous basis and in parallel with the other project planning processes.

A good quality planning process starts with a clear definition of the goals of the project. What is the product or deliverable likely to achieve? What does the product look like? What functions will it perform? How do you evaluate customer satisfaction? What determines the success of a project?

Answering these questions will help you in identifying and defining quality goals. It will also allow you to discuss the approach and plans required to accomplish those goals. This includes measuring the risks to success, setting high standards, documenting everything, and defining the methods and tests to attain, control, forecast and validate success. You should make sure that you include quality management tasks in the project plan and delegate the tasks to work groups and/or individuals who will report and track quality metrics.

For example, the desired management quality may need cost or schedule adjustments, or the desired product quality may need a detailed risk analysis of an identified problem. Before the ISO 9000 Series was developed, the activities mentioned here as quality planning were broadly considered as part of quality assurance.

Inputs to quality planning:

Quality policy: Quality policy refers to the overall intentions and direction of an organisation pertaining to quality, as formally expressed by the top management.

Scope statement: The scope statement comprises the key objectives of the project that are needed by different stakeholders.

Product description: It includes the details of technical issues and other concerns which may influence quality planning.

Standards and regulations: The project management team must acknowledge all the relevant standards or regulations that may influence the project.

Tools and techniques to quality planning

Benefit/cost analysis: The quality planning process should acknowledge benefit/cost trade-offs. The benefits should cover higher productivity, lower cost and high customer satisfaction.

Benchmarking: As per this process, we compare actual or planned project practices with the practices of other projects to produce ideas for improvement and to find a suitable standard to measure performance.

Flowcharting: It is a diagram that depicts how different elements of a system relate to each other.

Design of experiments: It is an analytical technique that helps in identifying which variables affect the overall income the most.



Outcomes from quality planning

Quality management plans: They provide input to the overall project plan and must deal with quality control, quality assurance and quality improvement for the project.

Operational definitions: It particularly explains what something is, and how it is measured by the quality control process.

Checklists: It is a structured tool that helps in verifying if a set of required steps has been performed.

Inputs to other processes: The quality planning process may discover a need for further activity in some other area.

Step 2: Quality assurance

Quality assurance means all the systematic and planned activities executed within the quality system give confidence that the project will meet the applicable quality standards. It also comprises the examination of data at stations and centres to validate that the data are in line with the quality system goals, and to identify errors so that required action can be taken in time.

A quality system should incorporate procedures that allow feedback into the measurement and quality control process to avoid the errors from recurring. We can apply quality assurance in real-time post measurement, and can also feed into the quality control process for the subsequent process of a quality system. However, it usually tends to operate in non-real time.

Quality assurance tests make use of a system of metrics to decide whether or not the quality plan is progressing in an acceptable manner. We can measure project quality with customer satisfaction effectively by using both qualitative and quantitative metrics. These tests or quality audits assist us in forecasting and verifying the accomplishment of goals and identifying need for corrective action.

Additionally, quality assurance tests also help us in mapping quality metrics to quality goals, thereby allowing us to report—on the status of quality—at periodic project review meetings.

In most cases, quality assurance is provided by a quality assurance department or similarly titled organisational unit, but it is not mandatory that only these departments perform this function. Quality assurance may be provided to the team managing the project. It may also be provided to the management of the performing organisation (internal quality assurance) or it may be provided to the customer and others not actively involved in the work of the project (external quality assurance).

Inputs to quality assurance

Apart from the quality management plans and operational definitions discussed under inputs to quality planning, the inputs to quality assurance include the following:

Results of quality control measurements: These are the records of quality control testing and measurement for comparison and analysis.

Tools and techniques for quality assurance



Quality audit: It is a structured review of other quality management activities. Quality audit may be carried out by properly trained in-house auditors or by third parties such as quality system registration agencies.

Outputs from quality assurance

Quality improvement: It includes all the actions taken for increasing the effectiveness and efficiency of the project to provide added benefits to the project stakeholders.

Step 3: Quality control

Quality control involves scrutinising particular project results to determine if they abide by relevant quality standards and identifying ways to remove causes of unsatisfactory results. Quality Control (QC) refers to a system of routine technical activities that assist in measuring and controlling the quality of the project as it is being developed. This includes identifying, analysing and correcting problems.

While quality assurance is provided prior to problem identification, quality control is reactionary and occurs after the identification of a problem. Quality control examines specific project results and determines conformity with relevant standards. In addition to this, it also identifies risk factors in a project, their mitigation and looks for ways to avoid and eliminate unsatisfactory performance.

The quality control system:

- provides routine and consistent checks to make sure that data is reliable, correct and complete;
- identifies and deals with errors and omissions; and
- documents and collects inventory material, and records all quality control activities.

Quality control activities comprise general methods such as accuracy tests on data acquisition and calculations and the use of accepted standardised procedures for emission calculations, measurements, estimating uncertainties, recording information and reporting. Higher-tier quality control activities comprise technical reviews of source categories, activities and emission factor data and methods.

Quality control implies working to a set standard of quality which is achievable and which has a ready market. Hence, quality control means adherence to a standard or prevention of a change from the set standard. In general, this is essential because when there is an acceptable quality, a manager must ensure that there is no deterioration from the standard.

However, in a changing world, one is often faced with the fact that the quality which is acceptable today by the customer may not be acceptable to him a year later. Therefore, there is need for a breakthrough, (creation of change) for improving the existing standards. Thus, preventing change (control) and creating change (breakthrough) are two important functions of quality management. Unfortunately, a large number of managers simply have no time for breakthrough because they are obsessed with day-to-day problems of keeping controls at the existing levels.

The quality control process includes:

- formulating the quality policy;
- setting the specifications or standards based on customers' preference,
- cost and profit;



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- selecting an inspection plan and setting up a procedure for checking;
- detecting deviations from set standards of specifications;
- taking corrective actions or essential changes to achieve those standards;
- taking decisions on salvage method, that is, to decide how the faulty parts will be disposed of, entire scrap or rework;
- coordination of problems concerning quality;
- developing quality awareness, both, within and outside the organisation; and
- developing procedures for good buyer-supplier relations.

The project management team should have practical knowledge of statistical quality control, specifically sampling and probability, to help them assess quality control outputs.

Inputs to quality control

The inputs to quality control include the work results, quality management plans, operational definitions and checklists. Apart from work results, the explanation of other inputs is discussed above.

Work results: It includes, both, process results and product results. In order to get the work results, information about the planned and actual results should be available.

Tools and techniques for quality control

Inspection: It includes activities such as measuring, examining and testing undertaken to determine whether the results conform to the requirements.

Control charts: Control charts are a graphic display of the results, over time, of a process. These are prepared to determine the variation between actual and planned outputs.

Pareto diagrams: A Pareto diagram is a histogram, ordered by frequency of occurrence that shows how many results were generated by the type or category of identified cause.

Statistical sampling: It determines the number of items to sample. For example, selecting 10 paintings at random out of a list of 100.

Flowcharting: It is used in quality control to help analyse how problems occur.

Trend analysis: It involves using mathematical techniques to forecast the future outcomes based on historical results.

Outcomes from quality control

Apart from the quality improvements, the following are the key outputs from quality control:

Acceptance decisions: The items inspected will be either accepted or rejected.

Rework: Rework is action taken to bring a defective or non-conforming item into compliance with requirements or specifications.

Completed checklists: It should become part of the project's records



Process adjustments: It includes the key preventive and corrective actions taken as a project control measure.

Value Engineering

Closely related to the concept of quality is the concept of value engineering.

Value engineering is the systematic application of recognised techniques that identify the function of a product, process or services, accomplishes a monetary value of that function, brings out areas of avoidable costs and provides the essential functions consistently at the lowest overall cost. In short, it tries to improve function or reduce cost or achieve both without compensating quality, reliability and safety.

The 'avoidable costs' depict that cost which does not provide desirable function, or quality, or life or appearance to the product.

So, what is cost?

Cost is the amount that we generally pay. It is the total sum of money spent on materials, labour, overheads, profit, etc., to manufacture a product.

Then, what is value?

Value is something that a customer believes the product is worth to meet his minimum specific requirements. It refers to the lowest price that a customer must pay to get a function or service.

A product is said to be some value if it has suitable performance and cost.

Value is thus defined as,

$$\text{Value (V)} = \frac{\text{Performance (P)}}{\text{Cost (C)}}$$

The value engineering process

A prime feature of value engineering (VE) is that a design or initial solution is usually in place before undertaking the process. This initial design is used right through the value engineering process to create optional approaches. The process of actually performing value engineering occurs in various circumstances.

In the US, in a 40-hour workshop, a design that is prepared by the client's design team is re-engineered by some other independent team. This team spends a week just looking at the original design in a workshop environment and makes its suggestions about alternative solutions.

There is usually only one 40-hour workshop on any particular project. In Europe, this 40-hour workshop has been modified into a two-day session with the original design team. The members of the design team meet to specifically challenge the team's own design and seek viable options.

This workshop is typically repeated during different phases of the design process and can extend onto the construction phase, in which the capability of the contractor can be exploited to pursue substitutes to be used in the particular design solution.



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Similar to the value management process described above, we can also implement value engineering through a five-stage plan, in which each stage is aligned to a generic problem-solving procedure.

Stages in value engineering

Information: An information gathering process in which the prime focus is on the client's business drivers for the project. Special relevance is given to the use of facilitated workshops.

Speculation: Creative-thinking techniques are put into use to generate alternative ways to provide the business drivers identified in stage 1.

Evaluation: The solutions generated are evaluated for their viability and cost. Ideas are combined and consolidated to generate a list of some five or six ideas that are worthy of further consideration.

Development: The ideas generated and shortlisted in the previous stage are developed in detail, ensuring that all the interfaces with the client's business are considered.

Recommendation/implementation: The most appropriate solution is identified and a formal recommendation is made to the client for implementation.

Scope of applying VE in a project

Projects aim to produce some product, process, services or others. A good number of these are concerned with building plants to produce goods and services to satisfy the human wants and needs. The typical life cycle of such projects includes project selection, planning, execution and closure.

VE may be applied in all the phases of project life cycle. However, the saving potential of a phase decreases as we move down the phases of a project. When the saving potential decreases, the cost to implement the changes to reap the benefit increases over the phases. As a result, the net saving decreases over the phases. This is why one should embark on VE as early as possible in a project.

Some important areas of VE application in various phases of a project include the following:

Project selection phase

- Listing of all potential projects.
- Shortlisting potentially good projects on technical and financial consideration.
- Market analysis: Product-mix, demand and supply, existing and future competition, etc.
- Technical analysis: Plant capacity, product mix, plant location, process selection, selection of equipment including specification, requirement of auxiliary and utilities, design and engineering of fabrication and civil work, etc.
- Financial analysis: Estimation of capital and operating cost, profitability analysis and sensitivity and risk analysis.

Planning and execution phase

Procurement of material and equipment accounts for over 50% of project cost. VE can play a useful role in cutting the cost and lead time.

Some VE approaches suggested by Miles to improve procurement are summarised below:



- Analyse the function performed by a part, particle or specification. If any of these contributes little or nothing, the designer should reconsider.
- Aesthetic function may be performed by various alternatives. Select the alternative performing the desired aesthetic at the least cost.
- Avoid extra cost needed to make the material ready to use.
- Analyse the different products performing the needed functions. Choose the product that costs the least.
- Compare the function provided by the buyer's process with the desired function. Select the process offering the closest fit.
- Identify the parts performing the needed function and the parts supporting the primary parts. Cut down the cost of supporting parts.

Benefits of VE in projects

Value engineering is a systematic process that aims to achieve the required functions at the least cost. Unlike value management, it tries to provide an optimal answer to a design problem. Value engineering advances from an initial solution to offer function-oriented alternatives created in an unconstrained, creative environment. There are also some other benefits of application of VE in projects as it:

- reduces the complexity of the product,
- achieves standardisation of components,
- ensures improvement in functional aspects of the product,
- ensures improvement in job design and safety,
- ensures improved maintainability (serviceability), and
- provides a robust design.

Summary

In this segment, you learned that:

- quality is a state in which value entitlement is realised for the customer and the provider in every aspect of the business relationship;
- the basic processes of project quality management include quality planning, quality assurance and quality control; and
- value engineering is a cost effective approach which improves the performance without disturbing the function(s) and without compromising on quality, reliability, etc.



Deliverables of Project Management

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1. Objectives
2. Introduction
3. Deliverables of Project Management
4. Establishing Checks and Balances
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe what a deliverable is
- Identify the deliverables of project management
- Describe the checks and balances of project management

Introduction

In the previous segment we learned about the concept of quality management and value engineering. In this segment, we will learn about the deliverables of a project, which conform to the quality requirements and offer a value proposition to the end customer

So what are deliverables?

Deliverable as an adjective describes something, which can be delivered.

This is a common term heard in the project management sphere. It is any tangible or intangible object—produced as a result of a project—that is intended to be delivered upon the completion of that project.

For a Web application project, the final product—Website up and running—is the deliverable. There may also be other deliverables associated with it such as the source code of the project, help documents, training materials and so on.

In our real-world example of a typical wedding, the wedding planner's catering arrangement is one of the project deliverables. Completion of the wedding ceremonies without any problems or issues may be a key deliverable for an event manager.

In a practical scenario, often times, people who are working do not know what they are producing. Whether the produced item will be useful or not could also be a question. The project's goals, initial scope, project stakeholders and end users are all determined during the initial stage of the project, either by the *Sponsor* or by his team or the *Project Manager*. The *Project Manager* has to define the deliverables that the project will produce. Even this is expected from the *Project Sponsor*.

To identify the project management deliverables, the *Project Manager* must try to understand the following:



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- What is the purpose and goal of the project?
- What does the client or stakeholder want at the end of the project?
- Are there various elements that must be approved within the project?
- Is the project goal tangible or intangible?
- Will the project require many levels of work prior to completion?

During the project initiation meetings, the expected deliverable from the task completion should be made clear, based on the plan when the tasks are assigned to the resources. Sometimes, completion of multiple tasks leads to one deliverable. Therefore, it must be carefully planned.

Deliverables of Project Management

Making a list of all deliverables expected from the *stakeholders* and the project is very important for the *Project Manager*. He/she needs to maintain a project deliverables list and track its status. Proper identification and tracking helps the *Project Manager* to plan appropriately.

The project deliverables again can be classified as those that require specific clients for approval and acceptance and those that are required by the organisation.

Some of the typical project deliverables include:

- source code of the technical project,
- design document of the project,
- deployment files of an application or project,
- training documents,
- quality validated results or testing results, and
- the completed product.

In the project life cycle phases, it is important to complete deliverables within that phase and get the approval of the stakeholders to move forward. Project management plan, scope plan and risk plan are also deliverables, which have to be created by the *Project Manager* either by himself or through his supporting team.

Establishing checks and balances

At this point, it is important to understand the checks and balances that facilitate high quality project deliverables.

Ensuring that a project is completed on time and within budget is a challenge. With the competing demands for resources and capital, projects have many internal and external forces that can contribute to their failure.

It takes a strong sense of direction, efficiency and leadership to keep the project on track. It is the responsibility of the *Sponsor* and the *Project Manager* to ensure that certain checks are in place to avoid such failures.

Some of these checks are as follows:

- The *Sponsor* has to spend extra time upfront to ensure that the business case is adequately defined and clear.
- It is possible that certain decisions about the project might have been taken up in haste. The *Project Manager* might be blamed later for an incomplete project or the failure of project. He



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should do a proper analysis before the commencement of the project and bring up issues and challenges upfront to the *Sponsor*.

- To complete project on time, the project team might overlook completion of certain functionalities. The *Project Manager* needs to clearly set guidelines and processes in place and ensure that such mistakes do not happen without his and the key stakeholders' (especially business *sponsor*) approval. This is done with the help of regular meetings and by adopting the 'always inform before judging' process.
- The *Project Manager* has to build contingency (buffer) in the schedule and the budget. Doing this gives more flexibility to the project in case certain risk-response plans have to be worked out. Even having certain backup resources would help in critical projects.
- It must be ensured that proper reporting capabilities such as generation of automatic reports, a process to receive status reports, etc. exists in the project. Automating many of these activities helps the *Project Manager* save time and focus on important issues.
- For better quality project deliverables, it must be ensured that the support of the right quality team exists. If the quality team is not present, other quality management techniques must be used.

The most important aspect of having checks and balances in a project is to have forethought on what areas one would like to focus upon. It can be thought from a risk angle, quality angle and a scope-schedule-budget angle.

Based on this, checks and balances must be prepared to ensure that there is minimal deviation from the terms agreed upon with the client.

Before moving from one phase of a project to the next, exit and entry criteria has to be adhered to. Team/Group leads should be made aware of such criteria. For instance, in a software project life cycle, the 'build' phase will not be started until the 'design' phase is complete.

Entry and exit criteria for the 'build' phase includes:

- completion of design documents in the design phase,
- two rounds of review against the design documents,
- review and approval of the start of work by the client team architects,
- ensuring that the total number of defects in the design are less than 10 and their impact is not beyond its module,
- approval by the *Project Manager*, and
- ensuring that the design documents cover the entire scope of the project and that the design is frozen.

The 'design' team leads verify that their deliverables meet all the above criteria and 'build' team leads take up build activity after verification and receiving confirmation.

Similarly, there may be many checks and balances placed in the project by the *Project Manager* to ensure that the project smoothly transitions from one phase to the next. The *Project Manager* may even place checks for approving expenses by team members.

Summary

In this segment, you learned that:

- a deliverable is any tangible or intangible object—produced as a result of a project—that is intended to be delivered upon the completion of that project;



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- it is important to complete deliverables within a particular phase of a project and get the approval of the stakeholders before moving forward to the next phase; and
- checks and balances are necessary to ensure that the deliverables are produced within time and budget.



Project Management Information System (PMIS)

Table of Contents

1. Objectives
2. Introduction
3. Project Management Information System (PMIS)
4. Planning of PMIS
5. Design of PMIS
6. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe Project Management Information System (PMIS)
- Identify the key steps involved in the planning of PMIS
- Design an effective PMIS

Introduction

In the previous segment, we learned about the deliverables of project management and the various checks and balances necessary to ensure that the deliverables are produced within time and budget.

In this segment, we will focus on the concept of PMIS, its planning and design.

The success of timely implementation of projects depends on the availability of essential information at the appropriate time. The information is needed for taking various decisions during selection, planning, execution and closure of a project. Project Management Information System (PMIS) aims to provide relevant information on time, resulting in improved performance.

Traditionally, this system was manual, paper based and labour intensive. It was slow to respond and update. The advancement of computer and telecommunication technology have revolutionised this system.

The features of the present PMIS include:

- Speed: Processes speed up the creation of information within a blink of an eye.
- Capacity: Provides ability to process and store large amounts of data.
- Efficiency: Fewer people are needed to manage the system.
- Economy: Provides cost advantage over the manual system.
- Accuracy: Provides better accuracy as compared to the manual system.

PMIS

PMIS refers to the system tools and techniques used in project management to deliver information. Project managers use the techniques and tools to collect, combine and distribute information through electronic and manual means. PMIS is used by the upper and lower management to communicate with each other.



Project Management

It is a Management Information System (MIS) related to a project. A PMIS consists of people, equipment and procedures to collect, process, store, combine and communicate the needed information to users (*stakeholders*) for carrying out project management functions.

PMIS helps in planning, executing and closing the project management goals. Project managers use PMIS for the budget framework such as estimating costs at the time of planning the process. Furthermore, the PMIS is employed to build a specific schedule and classify the scope baseline.

The project management team collects information into one database while executing the project management goals. It is used to compare the baseline with the actual achievement of each activity, manage materials, collect financial data and keep a record for reporting purposes.

To understand this concept better, we should first understand the distinction between data and information.

Data

Data are values of qualitative or quantitative variables belonging to a set of items. They represent something in the real world, expressed as a number or a statement or a picture and act as an input to MIS.

Information

Information may be defined as (i) data within a context, (ii) data in a relationship or (iii) processed data. It does not convey much. But, when we say it denotes an age of a student of a class, it gets some meaning. Log of daily production figures in a year do not convey much. But when we calculate their average, standard deviation, range and trend, we get much more meaning out of them.

System

A system consists of a set of sub-systems/components that work together to achieve a common goal. It comprises of interacting or interdependent components forming an integrated whole. Figure 13 depicts a data processing system.



Figure 13: Data Processing System

Most systems share common characteristics, which include:

- structure, defined by components/elements and their composition;
- behaviour, which involves inputs, processing and outputs of material, energy, information or data;
- interconnectivity, whereby the various parts of a system have functional as well as structural relationships to each other; and
- some functions or groups of functions.

As shown in Figure 13, the data processing system has four stages.



Project Management

Input

In the input stage, data are collected and entered into the computer.

Data processing

Data is transformed into information using mathematical, statistical or other tools.

Output

Output consists of information, which is displayed or presented.

Storage

Finally, in the storage stage, data and information are maintained for later use.

Planning of PMIS

The success of a PMIS depends on its effective planning. The PMIS is used for many purposes by a *Project Manager*, which include budget estimation of costs, creating a schedule, defining the scope, etc. Hence, these should be considered while planning for PMIS.

The planning of PMIS includes:

- identifying the information needed,
- capturing data,
- processing data into information and storing it, and
- communicating information to the stakeholders.

Identifying the information needed

Identifying the information needed is necessary for improving the decision making and the structure of the PMIS. Information requirements of project stakeholders include the recipients of information, the type of information needed, which includes format, contents and level of details, the time the information is required and how (by what media) will it be communicated to them.

(i) Who needs the information?

The recipients of the information may include any stakeholder of a project. They are individuals or groups that have a claim or ownership rights or interests in a project and its activities that may have occurred in the past, the present, or which may occur in the future.

Let us find how a project management information system is used by different stakeholders.



Table 10: Different Stakeholders in a Project Management Information System

Primary stakeholders	Organisational Managers: Directors, general managers, functional managers, project managers and project team members. Customer suppliers and contractors. Local, state and central agencies, commissions and judiciary, legislative and executive organisations. Unions, shareholders, creditors and employees.
Secondary stakeholders	Social and political organisations Environmentalists Competitors Local communities and the general public Media, various institutions (schools, hospitals, etc.) Tourists Professional institutions

Recipients of information may be decided on a 'need to know' basis. The list needs to be reviewed periodically to weed out obsolete entries and to add new ones.

(ii) What type of information is needed?

The information need of various stakeholders differs in content and with respect to the time of need. A survey may be conducted to identify the specific information requirements of stakeholders related to the various decision areas in a project.

However, most projects require information on some or all of the following:

- scope,
- time,
- cost,
- quality,
- resource utilisation,
- procurements, and
- risk.



Project Management

A comprehensive list of information needed during the various phases of a project is as follows:

The Selection Phase

The selection phase includes the following components:

Organisational guidance and support information: Project manual, project methodologies, policies, procedures, etc.

Information from other projects: Performance and best practices, budget, schedule and technical performance measures.

Statutory approvals and clearances: Several approvals and certifications such as environmental, labour, site approvals, safety standards, use of explosives and for high pressures/critical process operations are required.

The Planning Phase

The planning phase comprises:

- project scope;
- project cost;
- details of activities, which include descriptions, time, cost, resource needs and precedence relationships;
- project schedule and budget;
- drawing and documents including specifications, civil and structural drawings and layouts and schematics; and
- communication, including correspondences, minutes of the meetings, reports and the communication scheduled.

The Execution and Control Phase

Following are the components of the execution and the control phase:

Project organization: The project organisation structure includes the various functions and responsible agencies for the project and the coordination between the same.

This includes:

- list of stakeholders;
- disciplines, departments and specialties involved in the project;
- procedure for project execution, critical execution steps, deliverables and checks;
- safety and environment, training and inspection, cost and disaster management;
- issue logs, which include various issues that arise during a project; and
- change logs, where changes in scope, cost and time schedule are to be logged and reported.

Project reporting: Project reporting comprises three phases, namely project scope, resource requirement and the closure phase.

Project scope

The scope of the project includes the following:



Project schedule and cost: Two methods (variance analysis and earned value methods) are used to provide information on cost and schedule. Performance information generated by the two methods includes cost variance, schedule variance, cost performance index, schedule performance index, months behind schedule and cost overrun.

Quality records: A project sets up a quality system to ensure that incoming items (materials, parts and equipment), construction work, erection of equipment, etc., conform to the design specifications. The quality system generates a list of items which are below par for taking appropriate actions by the *Project Manager*.

Project risk assessment: It should be carried out at the start of the project as well as during the execution of the project; covering anything noticed or identified that may affect the future project schedule/cost. It is important to draw plans to implement measures to deal with any potential risks. These plans are to be implemented along with the project plan and monitored to identify any deviations so that corrective action can be taken.

Resource requirement

The resource requirement can be classified under the following heads:

Manpower: Report on manpower requirement can be presented in a tabular/graphical form or a combination of both. Figure 13 shows the manpower requirement during various phases of a project

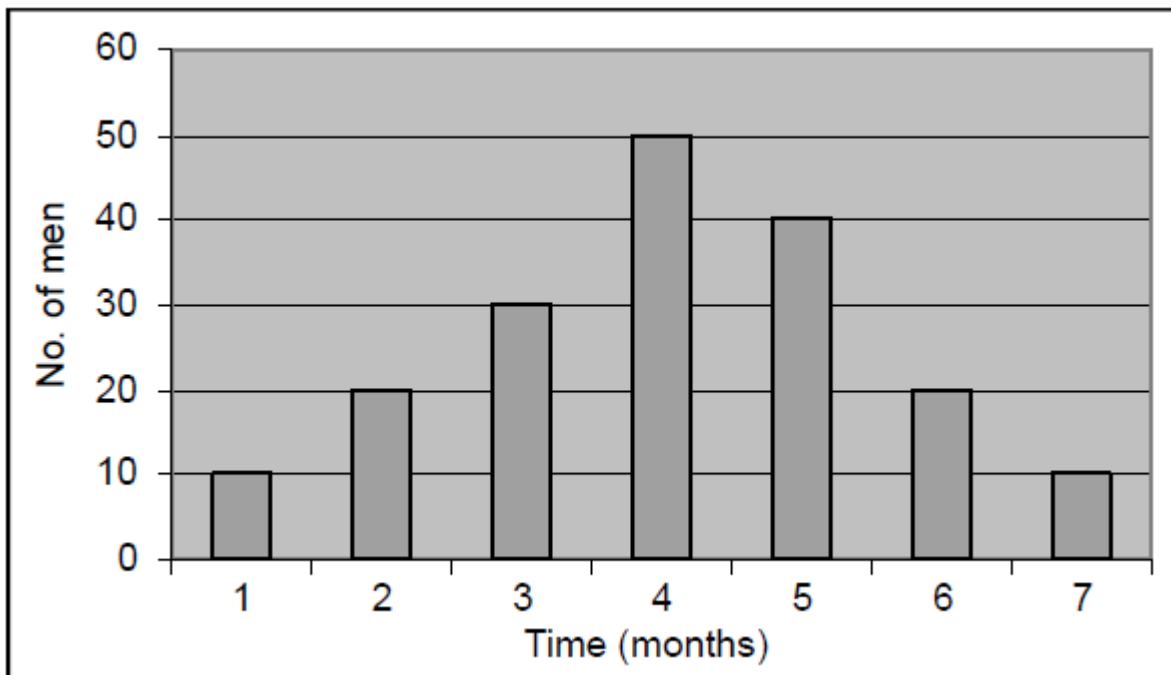


Figure 13: Manpower Needed for a Project

Materials, parts and equipment: A project needs a large number of materials, parts and equipment. Requirement of such materials, at different points of time, can be determined based on the network. With this input, material requirement planning is carried out for making these materials available on time.



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Many a times, proper linkage of requirements and availability are not ensured, resulting in time and cost over-run. At times, it affects the quality of work, as procurement of materials is hurried or substitute materials—not up to par—are procured.

Followup of the procurement process provides a list of materials, which need to be expedited (when materials are likely to arrive late) or de-expedite (when usage of materials is delayed due to any reasons).

Contracts: Besides other things, this function should include the sources of inputs/materials/services. This may prove a major source of advantage for the execution of the project and is kept at many organisations with secrecy. Many of the successful project managers can be seen to have such information.

Supplier/vendors performance—strengths and weaknesses: It includes the performance of various suppliers and vendors as was visible during the execution of the project.

Closure Phase

The closure phase consists of:

- review of achievement of goals,
- feedbacks on the performance, and
- lessons learnt.

(iii) When is the information needed?

Time of need of each stakeholder may be different. Some may need it on a daily, weekly or monthly basis.

Table 11: Format for Summarising the Information Need

Stakeholders	Information Description	How Often? (weekly/ fortnightly/ monthly)	Content and Format (summary/details, tabular/graphical)	Desired Media (report/meeting/electronic)

Capturing data

The term 'capturing data' is used to state a process of preparing and collecting data. The function of data collection is to attain information to maintain a record, to make decisions for vital issues and to pass information on to others.

Data can come from actual observation or from records. Data collected from records is known as secondary data. Data collected from direct observation is known as primary data.

It should be ensured that all relevant groups are represented in the data. A formal data collection process is essential as it makes certain that the gathered data are both defined and precise and that



subsequent decisions based on opinion embodied in the findings are valid. Data can possibly be arranged in a tabular form, data array or frequency distribution.

Processing data into information

An organisation—to achieve its aims—needs to process the data collected into meaningful information. It should be presented in its most useful formats. Data must be processed in a context to give it meaning. Data is transformed into information using mathematical, statistical, or other tools including computer software. Information can be stored in electronic form or hard copies represented in the most useful form.

Communicating information to stakeholders

Communication is the process by which information is exchanged between individuals through a common system of symbols, signs or behaviour.

To show the importance of communication, Cleland quotes the following statements:

- Peter Drucker states that the ability to communicate, heads the list of criteria for success.
- Harvard Business Review reports that the ability to communicate was at the top among the 22 personnel attributes considered for promotions.
- *A Project Manager* uses communication more than any other force to manage the project.

Channels of communication

The three most important means through which people generally communicate are:

- by actual physical touch, such as a pat on the shoulder, a pat on the back or the ritualistic expression of the handshake;
- by visible movement of some portion of their bodies such as the pointing of a finger, the wink of an eye, a smile, a nod or a grimace; and
- with the use of symbols, spoken or portrayed, which encompass some meaning based on experience.

The first two are referred to as non-verbal and the last one is referred to as the verbal channel of communication.

Some examples of means and channels of communication include plans, policies, procedures, objectives, goals, strategies, organisation structure, linear responsibility chart, meetings, letters, telephone calls, e-mails and project meetings.

Some common means of communicating information

There are a number of traditional as well as emerging methods of communicating information. The project manager may choose any combination of these. Some of the popular means of communication are discussed below.

Traditional methods of communication

Even with the arrival of modern methods of communication, some traditional methods are still more prevalent for the purpose of communication. Let us discuss a few of them.



Written reports: Written reports are age old methods to communicate project information in many organisations. These reports are generated at various levels of details and use tables, charts and graphs.

It should be borne in mind that written reports cost money in preparing, distributing, using and archiving. Longer is the report, more is the cost. A report should normally not exceed 5-10 pages.

To minimize cost, standard report formats may be defined that focus on key metrics by which projects should ideally be tracked and measured. Usually, distribution of these reports is displayed in a matrix. Column in a matrix shows the authorised recipients and a row shows each category of the report. Entry in the square at each intersection shows the recipient of a particular document at a particular frequency. If reports are made available over a network, appropriate security measures should be taken.

Face-to-face meeting: In face-to-face meetings, all participants sit in a room and participate in the agenda. The opinions about this form of communication are aptly described as, 'this is the most active form of communication and decision making' or 'this is the most ineffective use of time; an occasion, where the minutes are kept but hours are lost'. The difference is made by the way the meetings are conducted.

Emerging Methods of Communication

The emerging methods of communication or the modern-day methods of communication can be used where the traditional methods are not found to be satisfactory. They also have greater accessibility. The medium of communication is such that you can communicate with a person or persons sitting at a far off place.

E-Mail: E-mail has the advantage of reaching quickly to many people at dispersed locations. However, it should be used carefully and thoughtfully. Project report can be distributed as an attachment to an e-mail.

Internet site: Internet site or a Website is related to the web pages that restrain content (media), that is, text, video, music, audio, images, etc. A Website is hosted on no less than one Web server, which is accessible via a network.

For instance, the Internet or a private local area network through an Internet address known as a Uniform Resource Locator (URL). All publicly available websites together constitute the World Wide Web (WWW). An organization can design an internet site to retrace and view project information by authorised persons.

Chat room: Chat room is the online meeting of participants on project issues. Under this method, opinions are exchanged by typing messages and viewing responses on the computer's screen. The session can be archived for future use. In this case, the party has to obtain lease connection from an authorised body such as Bharat Sanchar Nigam Ltd., India.

Video conferencing: Video conferencing is the conduct of a videoconference (also known as a video conference or videoteleconference) by a set of telecommunication technologies which allow two or more locations to communicate by simultaneous two-way video and audio transmissions.



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It has also been called 'visual collaboration' and is a type of groupware. Here, two parties exchange their visual/audio data simultaneously. Infrastructure needed includes a computer, high speed LAN/wan, Internet connection and video conferencing software.

Blog: A Blog is a white board or web space over the Internet. It is a portion of the national Web space, acquired through purchase. Only blog owner or writer is authorised to write his views on the blog and edit it. The blog can be read by the interested audience. They can write their comments on it. However; they cannot edit it. The blog writer can accept or reject the comments. Blog writing is similar to delivering a lecture. A blog can be a part of the internet (connecting to the outside world), intranet (covering the organisation) and extranet (covering the organisation and a business partner).

Every country has its own Web space. Organizations/individuals can purchase a part of the space.

The objective of blog writing may include:

- disseminating information, and
- sharing knowledge on an issue with the purpose of reinforcing or modifying it.

A *Project Manager* may use a blog to communicate his views on project-related issues such as safety, motivational package, resource usages, unforeseen situations and the likes.

Wiki: A Wiki consists of Web space over Internet/intranet/extranet, shared by an authorised group of users to discuss their common objective. Every user has the right to read, write, and edit others' opinion or view expressed on the Wiki.

A Wiki uses a technology that assists a user in reading, writing, and editing. Besides, it assists in keeping a complete record of history of the process and retrieving any record. It is similar to case study discussion in a class.

Any member can write his view on any subject on a Wiki. Other members can edit the writing and add their opinions. This process continues till a verified, true and accepted view emerges. Objective: This is used to arrive at a consensus on an issue or to create knowledge through a collaborative process.

Advantages of using a Wiki:

- A wiki user does not need to have any computing or web-related expertise to add, edit or delete a page.
- It creates an effective library of knowledge.
- An employee can share, edit and access work-related information/knowledge such as reports, best practices and documents.
- A *Project Manager* may use this method to build consensus on an issue or find an approach for doing a thing.

Selecting an appropriate communication model

The choice of an appropriate medium of communication depends on specific purposes—such as assessing commitment, building consensus, mediating a conflict and others—to be addressed. Schnabel quotes some guidelines to select an appropriate medium to suit the specific need to communicate information.

A portion of the guidelines are presented in Table 12.



Table 12: Media Choice Matrix

How well is the medium suited to?	Hard copy	Phone call	Voice mail	e-mail	Meeting	Website
Assessing commitment	3	2	3	3	1	3
Building consensus	3	2	3	3	1	3
Mediating a conflict	3	2	3	3	1	3
Resolving a misunderstanding	3	1	3	3	2	3
Addressing negative behaviour	3	2	3	2	1	3
Expressing support/Appreciation	1	2	2	1	2	3

Key: 1-Excellent 2-Adequate 3-Inappropriate

Design of PMIS

The following elements should be kept in view in the design, development and operation of a PMIS:

- The PMIS should support the full range of life cycle including project analysis and post project review.
- Enterprise guidance and project background information must be a part of PMIS.
- PMIS includes all information coming from various sources, including formal reports, informal sources, observations, project review meetings and questioning.
- The PMIS must interface with the larger organisational information system to permit smooth, well-organised interchange of information in support of organisational and project goals.
- Only relevant information should be a part of PMIS.

Design of the PMIS consists of four sub-systems:

- Capturing data from primary as well as secondary sources.
- Processing data into information.
- Storing data/information/reports.
- Distributing/communicating information.

Each sub-system, in turn, consists of the following components:

Hardware: A computer and its peripheral: Input, output, and storage device. It also includes communication equipment that facilitates fast transmission and reception of text, pictures, sound and animation in the form of electronic data.

Software: Sets of instructions that tell the computer how to take data in, how to process it, how to display information and how to store data and information.

People: It includes IT professionals and users, who analyse information needs, design and construct information system, write computer program and operate and maintain the system.

Procedure: Procedures include priorities in running different applications and the security measures to achieve optimal and secure operations of the system.



Project Management

Successful project management depends greatly on careful design, implementation and maintenance of an effective PMIS. When the planning stage has been meticulous and the design of the PMIS is robust and in keeping with the characteristics discussed above, an organisation can look forward to a PMIS that meets its stated requirements and ensures project success.

Summary

In this segment, you learned that:

- PMIS is a MIS comprising system tools and techniques that are used in project management to deliver information.
- A good PMIS must provide appropriate information about the key factors critical to the success of the project.
- The success of a PMIS depends on its effective planning.
- There are some key elements that must be ensured in the design, development and operation of a PMIS.



Creating a Work Breakdown Structure (WBS)

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1. Objectives
2. Introduction
3. Work Breakdown Structure (WBS)
4. How to create a WBS?
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe what a Work Breakdown Structure (WBS) is
- Describe how to create a WBS

Introduction

In the previous segment, we learned about the project management information software. In this segment, we will learn how to create a work breakdown structure.

A work breakdown structure is needed because the scope of work in the project has to be broken down into smaller pieces so that they can be managed better.

WBS

Breaking down of work into smaller pieces in project management is termed as Work Breakdown Structure, in short—WBS. It is one of most common term used during the early phases of a project such as initiation, planning and assignment of work.

Why do we need to breakdown tasks? What happens if they are not broken down?

There are multiple reasons for using a WBS in projects—some of these are listed below:

- Breaking down work ensures more accuracy and specifically helps to define and organise the scope of the total project. Each level of the structure breaks the project deliverables or objectives down to more specific and measurable chunks.
- Using WBS in projects helps in assigning responsibilities, resource allocation, monitoring and controlling the project. The WBS makes the deliverables more precise and concrete so that the project team knows exactly what has to be accomplished within each deliverable. This also allows for better estimation of cost, risk and time because one can work from the smaller tasks to the level of the entire project.
- It allows double checking of all the deliverable specifics with the stakeholders and making sure that nothing is missing/overlapping.
- WBS plays an important role in identifying project milestones.



- Last but not least, WBS illustrates the project scope so that stakeholders can have a better understanding of the same.

Following is an outline of a sample WBS:

Title	Task W1	W1.1	
		W1.2	W1.2.1
		W1.3	W1.3.1
		W1.4	W1.4.1
	Task W2	W2.1	W2.1.1
		W2.2	W2.2.1
		W2.3	W2.3.1
		W2.4	W2.4.1

How to create a WBS?

The first step in creating a WBS is to get a team and, possibly, also the key stakeholders, together in one room. The team possesses all the expertise, experience and creative thinking that will be needed to get down to the specifics of each deliverable. If the team does not have the expertise then subject matter experts—or the people who have worked on similar projects in the organization—would be required for help.

There is nothing written in stone about how to break down the scope of work. It varies from project to project and manager to manager.

Generally, the first level comprises the project title and the second level comprises the deliverables for the project. Once we have the first two levels in place, it is time to break down work (decomposition).

‘Decomposition’ is the act of breaking down deliverables into successively smaller chunks of work to be completed in order to achieve a level of work that can be both realistically managed by the *Project Manager* and completed within the given time frame by one or more team members. This level of breakdown and detail is called the ‘work package’.

Work packages are the lowest level of the WBS and are pieces of work that are specifically assigned to one person or a team of people. It is also the level at which the *Project Manager* has to monitor all project work.

Now, the biggest question is how specific and small can this work package be. It can be done by applying certain rules such as 8/80 or 8/40, that is, no work package should be less than 8 hours and more than 80 hours or less than 8 hours and more than 40 hours.

Does it mean that there is no work package less than 8 hours?



Project Management

The answer is, it depends on the type of project.

As a general observation, it is found that managing/tracking work less than 8 hours is tedious and time consuming for the *Project Manager*. However, in certain cases, it becomes important to measure work of even 30 minutes or less.

Most of this work is performed by either machines or automated tools and the *Project Manager* or the team member only observes the result at regular intervals.

The efficient breaking down of work helps the *Project Manager* to manage properly. It is one more step to a successful project.

If all the above requirements have been followed, the *Project Manager* will be able to use the work breakdown structure to arrive at the project work plan.

In this way, WBS plays an important role in the project life cycle and sets the tone for the execution phase of project management.

Summary

In this segment, you learned that:

- breaking down of work into smaller pieces in project management is termed as Work Breakdown structure;
- WBS serves multiple purposes and is, therefore, an important aspect of the project life cycle; and
- creating a WBS is not a static process, rather it is dynamic in nature and can accommodate the particular needs of a given project.



Executing a Project

Table of Contents

1. Objectives
2. Introduction
3. Project Execution
4. The Project Control Process
5. The Purpose of Project Execution and Control
6. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the project execution phase
- Describe the project control process
- Explain the purpose of project execution and control

Introduction

In the last segment, we learned about WBS and its importance in the project management life cycle. In this segment, we will focus on project execution and the project control process.

So, what does project execution mean?

An inexperienced person may interpret the term 'execution' to mean the personal consequence (or at least it feels that way) if their project fails.

To a *Project Manager* or a leader it signifies the 'Who, What, When and How' that guides all project activities. Sound project execution is a skill and also a knowledge area.

Technical skills and good people alone are not sufficient. Good project execution is possible when a project team fully understands its roles, responsibilities, deliverables and due dates and works together to complete the required work within the scheduled time.

Project teams show a collective morale and cohesion (or not) that is significantly influenced by how well their work processes, communication and interactions work together. All these are execution elements.

If confusion reigns, then productivity, quality and morale decline and schedules are missed. If this occurs, it is time to stop, step back and understand what is happening. Where is the problem? Is there a problem in the plan, or in the working? These are some of the important questions that must be asked.

During project execution, the *Project Manager* as to act like a manager, coordinator, well-wisher, motivator, coach and takes on other roles that are relevant to make the project successful.



Project Management

Performance measurement and evaluation are important aspects in any project management process. Performance measurement is the ongoing, regular collection of information for monitoring how a project, policy or strategy is doing.

On the other hand, evaluation involves comparisons of performance with the objectives of the information organisation in order to determine the change in performance over a given period or to see whether the change is in the right direction or not and if so then to what extent.

Project execution (or implementation) is the phase in which the plan designed in the prior phases is put into action. The purpose of project execution is to deliver the project's expected results (deliverable and other direct outputs). Typically, this is the longest phase of the project management lifecycle where most resources are applied.

During project execution, the execution team utilises all the schedules, procedures and templates that were prepared and anticipated during the previous phases. The completion of start-up activities ensures that all elements are in place and project is ready to take-off. Project execution converts the plan into a tangible asset, fulfilling a set of objectives. However, in real life, this transition from a plan to a tangible asset is not smooth.

In the standard division of project management discipline, this phase is known as 'project execution and control'; the term 'control' is included here because execution is not a blind implementation of what was written in advance but is a watchful process where doing things goes along with understanding what is being done, and re-doing it or doing it differently when the action does not fully respond to what it was meant for. This 'control' is an integral part of project management and is a necessary task of the *Project Manager*.

The task of a *Project Manager* is to periodically monitor the performance, calculate the deviations, and make adjustment in input, process, or plan so as to resolve the deviation and bring back the project to the desired state.

Project Execution

At the time a project moves into the execution phase, the project team and the necessary resources to carry out the project should be in place and ready to perform project activities. The project plan should also have been completed and base-lined by this time. The project team and, specifically, the *Project Manager's* focus now shifts from planning the project efforts to participating, observing and analysing the work being done.

Once all essentials are in place, the *Project Manager* plans to commence the project. He/she organises a meeting called the 'kick-off meeting or event' of all concerned, chiefly, to announce the start of the project. He/she takes the opportunity to present the main features of the project, project objectives, organisation and so on. The purpose of the meeting is to ensure that:

- team members understand the features of the project;
- team members understand the project objectives;
- all issues and apprehensions are clarified, so that each starts the project with a clear mind; and
- each member understands his/her role, authority and responsibility.



Project Manager's responsibilities

A *Project Manager's* responsibilities do not stop once the planning of the project is done. Because a project manager is responsible to the internal and external *Stakeholders*, the *Project Team*, vendors, executive management and so on, the visibility of the position is intensified. Many of these people will now expect to see and discuss the resulting deliverables that were so meticulously detailed in the planning phase.

As a *Project Manager*, keeping oneself from getting 'down in the weeds,' especially on large projects, will be important during project execution. This will allow the *Project Manager* to focus his/her attention on enabling the project plans, processes and managing the expectations of the customers as well as the *Stakeholders*.

During project execution, particular attention needs to be paid to keeping interested parties up to date with project status, dealing with procurement and contract administration issues, helping manage quality control and monitoring project risk.

While the processes to control many of these elements are discussed within the project control phase, it is, nevertheless, still important that the *Project Manager* be cognisant of the issues as the project is being performed. Daily interaction and feedback from team members, therefore, prove to be vital for project success.

Project team responsibilities

The *Project Team Members* are expected to assist in the management of the project as well; albeit, at a more functional level. The critical project management elements where the *Project Team* can provide assistance, include:

Performance monitoring: This includes implementing an execution plan to measure the actual performance as compared to planned performance. For example, the actual project schedules need to be reviewed periodically and compared to baseline schedules in order to discern if the project is performing according to the plan.

If the project is not performing according to baseline, steps need to be taken to get the project back on track. The same monitoring and analysing should take place on budgets, quality, risks, scope and so on.

Providing project status: While the *Project Manager* is responsible for relaying the project status to parties outside the *Project Team*, the latter is expected to report the status to the *Project Manager*. This includes communicating information, both, on a formal as well as an informal basis.

The issue of network, work to list and work order form

For execution of a large project, many a times, three types of network are prepared. These include:

Critical path network: Critical path networks highlight which tasks are critical for a project to stay on schedule and which can slide without affecting the completion date of the project. This allows efficient prioritisation of tasks. This shows only critical path activities and may be used by senior level executives.

Detailed network: This is the final network which is arrived at after the resource rescheduling process. It satisfies resource constraints and shows all activities including activities of the critical path.



Agency wise network: This network only shows the activities to be executed by a particular agency. It provides easy accounting for the executing agency.

Work-to List: Work-to list is a list prepared separately for each agency, after the resource scheduling process. It is a recommendation to the concerned managers of the sequence and timing of work to be done—a list to be worked upon. A work-to list is usually prepared to cover only a few weeks ahead. However, it may cover the details of longer periods as per need.

The Project Control Process

Control of the project is exercised through formal as well as informal processes exercised by the *Project Manager*, *Project Team* and the *Stakeholders*. The process of conducting reviews and monitoring reports exerts a degree of control over the project. This discussion will, however, focus on the formal processes of control established by the project plan. Because a project has a lot of associated uncertainties, project control is essential.

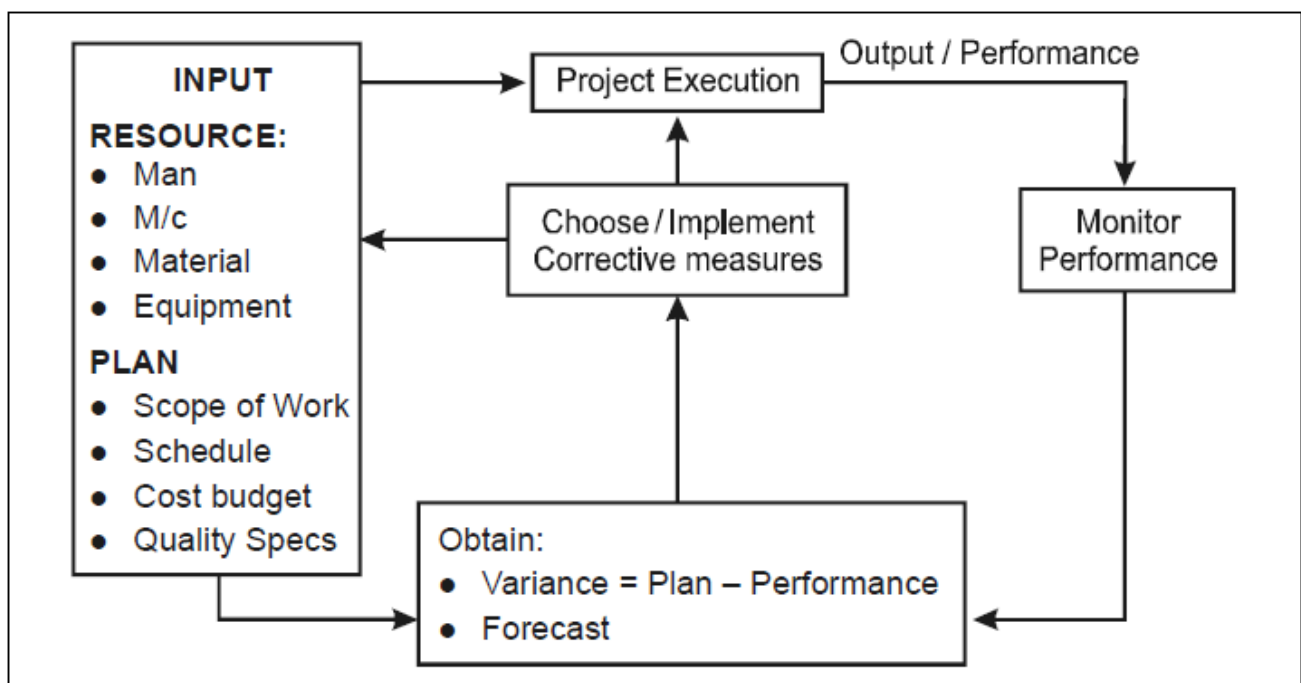


Figure 14: The System of Project Control

Identifying output and performance objective

There are three basic performance objectives/standards for a project which a manager is expected to control. These include time schedule (project duration), project cost and conformance to design specifications, which ensure quality.

Two of the commonly used methods to evaluate these objectives are variance analysis and the earned value method.



Performance monitoring for schedule and cost

Performance monitoring for schedule and cost tells who is to monitor the information and at what frequency. Normally, for a moderate size project, a separate project monitoring cell is created for collecting, analysing and reporting information to all concerned for the purpose of information and controlling the project. The task of the group may include:

- monitoring progress,
- calculating variance/earned value,
- preparing variance/earned value report and other reports and sending them to all concerned,
- presenting the report in the review meeting,
- compiling the measures arrived at in the meeting to deal with variances, and
- following up the implementation of suggested measures.

The frequency of monitoring depends on the nature, duration and importance of projects and the *Stakeholders'* need. The time between monitoring may vary from continuous to several days.

Cost and schedule performance measure – Earned Value Method (EVM)

The Earned Value Method (EVM) is a useful tool that allows the calculations of cost and schedule performance measures including cost variance, schedule variance and cost and time over-runs for a project. EVM uses the following parameters to calculate these measures:

Budgeted Cost of Work Schedule (BCWS): This is the budgeted cost of work scheduled up to the status date and is calculated as:

$$BCWS = \sum (\text{Budgeted cost of work/day} \times \text{Scheduled days of work up to status date for each activity})$$

Budgeted Cost of Work Actually Performed (BCWP) or Earned Value (EV): This is the budgeted cost of work actually accomplished up to the status date and is calculated as:

$$BCWP = \sum (\text{Budgeted cost of an activity} \times \text{proportion of it actually accomplished up to status date}).$$

Where,

$$\text{Proportion of an activity} = (\text{work completed in physical units}) / (\text{Total work planned in physical units})$$

Actual Cost of Work Performed (ACWP): This is the cumulative actual cost of work performed up to the status date and is calculated as:

$$ACWP = \sum (\text{Actual cost of work performed for each activity up to status date})$$

Using the above parameters, performance measures can be calculated as:

$$\text{Cost Variance} = BCWP - ACWP \dots (1)$$

A negative cost variance means that the project is spending more than it should.

$$\text{Schedule variance (cost)} = BCWP - BCWS \dots (2)$$

Negative schedule variance indicates that project is behind the schedule.



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Cost performance index (CPI) = $BCWP/ACWP$... (3)

CPI less than one, indicates that the project is spending more than the schedule and is unfavourable.

Schedule performance index (SPI) = $BCWP/BCWS$... (4)

SPI less than one suggests that the work performed is less than the schedule and is naturally unfavourable.

Figure 15 depicts the plot of BCWS versus time. This curve shows the cumulative spending planned for a project and is referred to as the baseline plan and is also referred to as the S-curve.

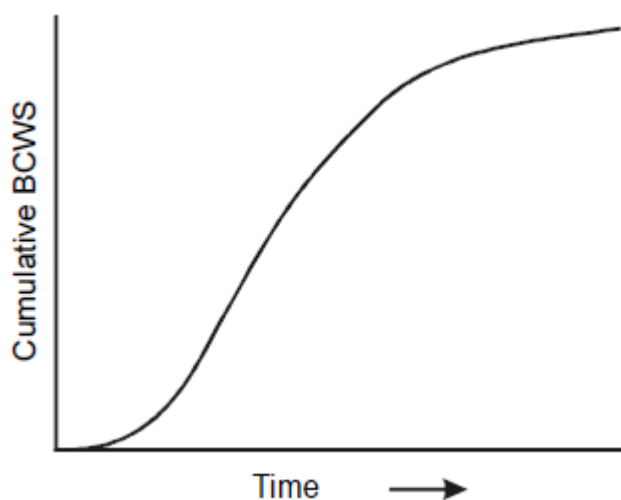


Figure 15: Budgeted Cost of Work Scheduled

Figure 16 depicts the plots of three spending BCWS, ACWP and BCWP for a typical project. It helps to calculate the cost as well as schedule variances, and hence, is quite useful in controlling the project.

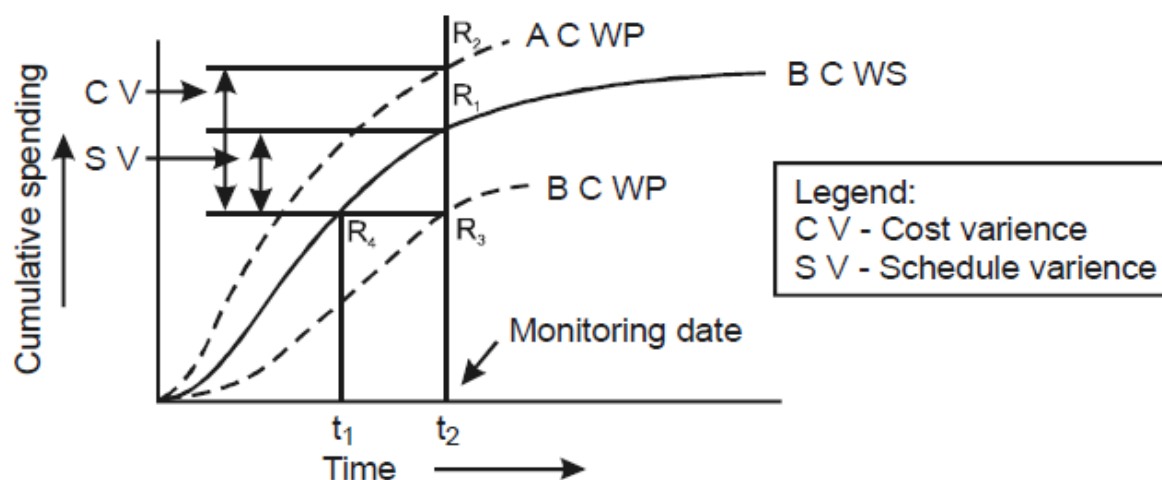


Figure 16: Spending Curves for BCWP, BCWS and ACWP



Variances and cost of overrun

The vertical distance between point R_2 and R_3 (that is, $ACWP - BCWP$) give the cost variance (in this case overrun).

$$\text{Cost of over-run (\%)} = \frac{ACWP - BCWP}{BCWP} \times 100$$

Schedule variance and months behind schedule

Point R_3 falls below point R_1 , indicating that the project is behind schedule. Schedule variance in monetary unit is given by vertical distance R_1R_3 (that is, $BCWS - BCWP$) and months behind is given by the horizontal distance from point R_3 on the $BCWP$ curve to R_4 on the $BCWS$ curve, which is equal to $(t_2 - t_1)$.

The above analysis of various spending ($BCWS$, $BCWP$ and $ACWP$) variances and their plots is carried out on an aggregate basis. Aggregate analysis simply shows whether a project is under control or not with respect to cost and time objective.

However, it does not pinpoint the specific actionable areas for bringing the project on track. For this reason, variance analysis needs to be conducted on an activity-to-activity basis, besides aggregate analysis.

Schedule performance measure using variance analysis

Variance is the deviation of actual value from the planned value of a parameter. Variance analysis is used to calculate the variances of parameters. Schedule Variance (time) is used to measure schedule performance of an activity in terms of time and is calculated as:

$$\text{Schedule Variance} = \text{Plan time} - \text{actual time}$$

Where, time can be measured in physical unit (days/months) or % complete.

The variance report may be presented in a tabular form or a graphical form. The data can be provided for the current period as well as on a cumulative basis for an activity. A graphical report may include GANTT chart, a milestone chart and a network diagram.

Here, the progress of any activity may be depicted by drawing another line or bar (usually in different colour) over the line or the bar indicating the activity in plan. Difference in the length of these two lines shows whether the activity is lagging, leading or on schedule.

Forecast: Once the current status of the project in terms of variances is known, the next task is to make a forecast about the project at completion. The forecast is an estimate of the future based on current progress and other factors (risks, resource availability and so on.)

Choose/implement corrective measures

The findings about variance and forecast are compiled in the form of a report. The report is circulated to all concerned and is discussed in a review meeting. Various options including re-planning, reallocating resources or changing the way the project is managed and so on are discussed to resolve the variances.



Of these, a set of options are chosen. Plans to implement these options are drawn and executing agencies are identified. The progress of such plans are monitored in the review meeting.

The Purpose of Project Execution and Control

The purpose of project execution and control is to develop the product or service that the project was commissioned to deliver. Typically, this is the longest phase of the project management lifecycle, where most resources are applied.

Project execution and control utilises all the plans, schedules, procedures, and templates that were prepared and anticipated during prior phases. Unanticipated events and situations will inevitably be encountered, and the *Project Manager* and the *Project Team* will be taxed to capacity to deal with them while minimising the impact on the project's CSSQ (Cost, Scope, Schedule and Quality).

The conclusion of the phase arrives when the product of the project is fully developed, tested, accepted, implemented and transitioned to the performing organisation. Accurate records need to be kept throughout this phase. They serve as input to the final phase—project closeout or project closure.

Summary

In this segment, you learned that:

- project execution (or implementation) is the phase in which the plan designed in the prior phases is put into action;
- control—corrective measures to resolve deviation between the planned and the actual—is an integral part of project management and is, therefore, included in the execution phase;
- two of the commonly used methods to evaluate variances of cost and schedule are variance analysis and the earned value method.



Reviews in a Project

Table of Contents

1. Objectives
2. Introduction
3. Review and its Types
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe what a review is
- Identify the various types of reviews

Introduction

In the last segment, we learned about project execution and the project control process. In this segment, we will focus on the importance of review and its various types.

So, what is a review?

A review is a formal or informal assessment or examination of something against its intended purpose of creation.

Review of deliverables in a project is done either formally or informally. Formal reviews are performed as per the process defined and accepted by all. Informal reviews are those wherein formal documentation can be skipped.

During preparation of a work plan, the *Project Manager* must allocate some time in the project schedule for project reviews. The *Project Manager* has to schedule some time to ask the following questions:

- What is the objective of the project?
- Does the outcome of this plan serve the project goal?
- Is this the best approach?
- Have we assigned the right resources?
- Are all the *Stakeholders* in agreement?
- Have we allocated enough budget?
- Is the break down of work appropriate?

The *Project Manager* has to get all deliverables of the project reviewed properly. The main deliverables of the project such as project plan, risk plan, issue plan, quality plan, scope plan and work plan, all have to be reviewed by subject matter experts to ensure that they support the goal, approach and feasibility of the project on ground. Post review, these plans have to be taken up with the *Stakeholders* for their approval.



The different types of reviews include:

- peer review,
- external project review,
- quality review, and
- stakeholder review.

Peer reviews

Peer reviews are conducted by colleagues or co-workers working in the same project. The *Project Manager* must ensure that proper templates, checklists, samples and trainings are provided for everyone to perform the peer review efficiently. It is possible that peers may not be well versed with the function of an item. In such cases the peer review is likely to be ineffective. Therefore, it is important to plan in advance.

External project review

This type of a review is performed by a person outside the project and sometimes outside the working group but by one who has knowledge or expertise in that particular field of work. When peer review fails or are found to be ineffective, external project reviews help.

It does not, however, mean that when peer review is performed, an external review is not necessary. It could still be required. External reviews also bring in a different perspective to the review process as they are not directly influenced by the *Project Stakeholders*.

Quality team review

Although the peer and external reviews are also quality reviews, a specific quality review is done by trained quality team members. The quality team brings in expertise and works outside the influence of the project or the *Project Manager*.

The review plan has to be prepared properly and shared with everyone. *Stakeholders* will be more interested to know the results of the quality review done by an external team.

Stakeholder review

As the name suggests, a *Stakeholder* review is done by any other stakeholders than those mentioned above. It could be the *Client*, *Project Manager*, client representative, *Sponsor* and so on.

The *Project Manager* has to decide which review is suitable for a given project and in which phase. These review schedules have to be documented in the work plan and the reviewer's time has to be blocked beforehand.

A generic review cycle involves creating the review plan, performing the review, recording the review and acting on the outcome of the review.

Figure 17 shows a generic review cycle.

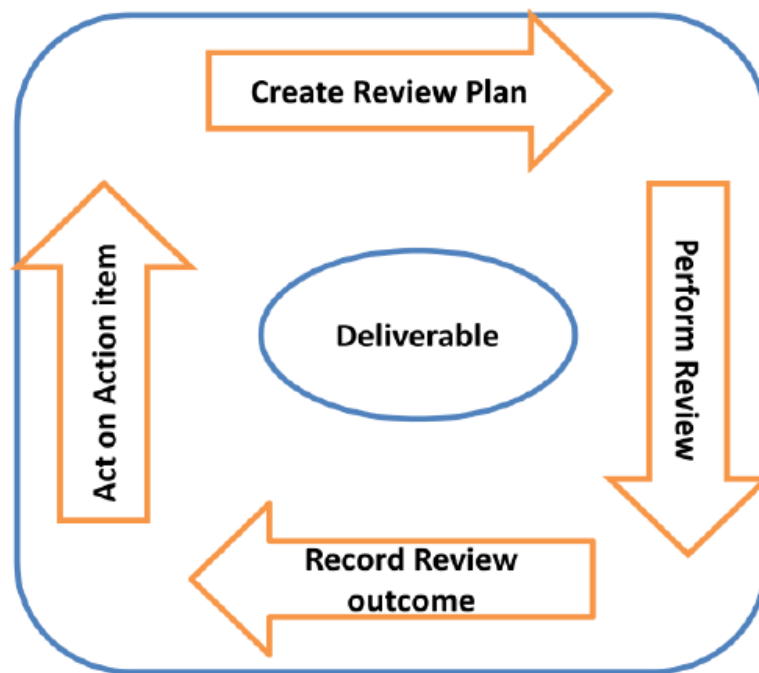


Figure 17: The Review Cycle

Even the project plan prepared by the project Manager or his team has to be reviewed to ensure its feasibility. If the *Project Manager* has prepared the plan himself, he/she has to find a peer *Project Manager* or a quality assurance team member to ensure the effectiveness of the review.

At times, the project plan may have some gaps which can be found earlier than later—by following the above process—thereby preventing cost implications to the project. The comments provided by such reviews need to be documented and action must be taken to ensure that they are satisfactorily closed.

In certain type of projects—where documentation is very critical and people who create these have to be assessed for their productivity and quality of work—data points such as the number of review comments, criticality of comments and size of the document, all could help.

Ideally, the *Project Manager* has to cap-in at least 10-15% of effort towards such reviews. Proper reviews done in initial stage can be very helpful. It is observed that the cost of fixing issues later in the project is very high compared to their identification and fixing in the early stages. The famous proverb, 'A stitch in time saves nine' is relevant here.

Let us take the example of making a bolt with a head of six sides and the length of 2 cm. If the person is not experienced on the job and has set his machine improperly, whereby it produces a length of 2.1 cm instead of 2 cm and is running for $\frac{1}{2}$ hour, we can imagine what would be the loss.

If the *Project Manager* understands the skill of the resource and—in the plan—makes sure that before making the bolts, the machine is verified by another expert or his peer, it can reduce the loss of effort and wastage.

Similarly, in the software industry, if the code written by one of the developers is reviewed properly by peers or leads, it reduces the defects on moving to the final stage of the project. The cost of



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fixing a defect—when found at the last minute—is high. Such defects reduce the confidence of the client—if found by them.

Hence, reviews serve an important purpose in the project life cycle and go a long way in ensuring that a fresh pair of eyes—whether it is that of a peer or any of the other entities discussed so far—are able to detect any potential flaws in the project at an early stage where corrective measures can be taken as against finding them at a later stage when it is much too late.

Summary

In this segment, you learned that:

- a review is a formal or informal assessment or examination of something against its intended purpose of creation; and
- project reviews can be performed by peers, the quality team, an external entity or the *stakeholders*.